

New developments and applications of human organoids

Amanda Andersson-Rolf¹  & Hans Clevers^{2,3,4,5} 

Abstract

Organoid technology offers unique opportunities for studying human biology and disease in vitro. Organoids are self-organizing 3D structures, derived from pluripotent or tissue-resident stem cells that recapitulate key aspects of primary tissues. Compared with classical cell lines, organoids provide distinct advantages. They can be derived from both healthy tissues and diseased tissues, enabling the investigation of disease mechanisms and the development of personalized therapies, and they better recapitulate the cellular heterogeneity of the native tissue, allowing for better modelling of human (patho)physiology. Although current organoids have provided valuable insights, these insights are inherently reductionist and do not fully capture the complexity of human tissues. The research field is, therefore, moving towards next-generation models that more accurately represent the intricate cellular interactions, tissue architecture and microenvironmental cues that underlie human biology and disease. In this Review, we outline the limitations and challenges of current organoid systems, highlight recent advances aimed at increasing their complexity, and discuss innovations that support their translation into clinical applications. The focus is on human tissue stem cell-derived organoids, with comparisons to pluripotent stem cell-derived organoids where relevant. We conclude by identifying key factors and remaining challenges for developing the next generation of organoids.

Sections

[Introduction](#)[Limitations of TSC-derived organoids](#)[Increasing complexity](#)[Disease modelling](#)[Studying tissue regeneration](#)[Organoids in drug discovery and development](#)[Conclusions and perspectives](#)

¹Department of Medical Biochemistry and Biophysics, Karolinska Institutet, Solna, Sweden. ²Hubrecht Institute, Utrecht, the Netherlands. ³Royal Netherlands Academy of Arts and Sciences (KNAW), Utrecht, the Netherlands.

⁴Onco Institute, Utrecht, the Netherlands. ⁵University Medical Centre Utrecht, Utrecht, the Netherlands.

 e-mail: amanda.andersson.rolf@ki.se; h.clevers@hubrecht.eu

Introduction

Model systems are indispensable tools for studying development and disease. Discoveries made in non-human animal models can often be extrapolated to humans, owing to evolutionary interspecies conserved mechanisms^{1,2}. However, animal models cannot fully recapitulate the complexities of human pathophysiology. Approximately 1% of human genes lack identifiable mouse homologues, and many biological processes, such as brain development³ and metabolism³, exhibit substantially interspecies differences. Dissimilarities are also apparent when modelling diseases. For example, murine tumours have a substantially lower mutational burden than human tumours, suggesting key differences in disease progression⁴. These variations can lead to discrepancies in drug metabolism and toxicity^{5,6}. Similarly, multiple pathogenic microbes do not infect animals or do not cause disease after infection^{7–12}. Therefore, advancing our understanding of human (patho)physiology requires the development of human-relevant model systems.

Organoids, which can be derived from pluripotent stem cells (PSCs) or tissue-resident stem cells, named hereafter as tissue stem cells (TSCs), are the latest addition to the repertoire of human model systems (Fig. 1). The meaning of the term ‘organoid’ has changed over time. When first coined in the 1950s, organoid referred to any type of organelle^{13,14}. A decade later, the term was used to describe a certain type of tumour^{15–17}, and thereafter, it was used to define cellular structures or cell aggregates generated via dissociation and re-aggregation experiments^{18–20}. Bissell then championed the use of Matrigel, an extracellular matrix (ECM)-like preparation to allow the formation of polarized, 3D mammary epithelial structures from an immortalized breast cell lines²¹. Compared with today’s organoids, these early 3D cultures lacked the cellular and genomic heterogeneity of the original tumour, representing an important limitation of early 3D culture approaches.

Self-organization is a defining feature of organoids. The ability of PSCs to give rise to self-organizing structures was first observed by the group of Sasai, when aggregates of mouse embryonic stem cells (mESCs) self-organized into 3D polarized cerebral cortical tissue²². Further optimization of the tissue medium and the addition of Matrigel allowed them to generate self-organizing optic cup structures in 3D. Starting from aggregates of mESCs, the retinal epithelium spontaneously formed hemispherical epithelial vesicles resembling retinal structures during development²³. Soon after, it was shown that TSCs also have the capability to self-organize into 3D structures in vitro^{24,25}. These studies have reported that a single adult intestinal stem cell can proliferate long-term in vitro and self-organize into 3D structures containing all common intestinal cell types and resembling intestinal crypts and villi^{24–26}. Together, these studies have demonstrated the remarkable capacity of stem cells to give rise to self-organizing 3D structures, and laid the foundation for modern organoid technology. Moreover, these studies have highlighted the unique ability of organoids to largely preserve the cellular heterogeneity or the organ of origin. In the years that followed, organoids from healthy and diseased tissues were generated, including organs such as the brain²⁷, liver^{28,29} and pancreas³⁰. Compared with proliferative tissues such as the intestine, these quiescent organs are characterized by low rates of cellular turnover and harbour comparatively few actively dividing stem cells, making organoid derivation more challenging. The first TSC-derived organoid from the brain was only recently established³¹.

Research over the past 15 years has highlighted the complementary nature of TSC-derived and PSC-derived organoids. TSC-derived organoids excel at modelling the epithelial compartments from fetal and

adult organs, particularly from the endoderm. TSC-derived organoids are derived from primary tissue and can be expanded long-term, whereas PSC-derived organoids are generated via guided differentiation of PSCs and result in a static end product; that is, upon reaching the end point of the developmental trajectory, they cannot be further expanded (Fig. 1). PSCs have the ability to model the earliest stages of mesoderm and neuroectoderm development (where proliferative stem cells and/or progenitors are present). Consequently, heart³², kidney³³ and brain²⁷ organoids with more complex structures have been established using PSCs. This complementary relationship underscores the importance of utilizing diverse organoid models to capture the full spectrum of human biology.

Recent research has also revealed limitations in the ability of some organoid types to fully recapitulate the native tissue architecture and heterogeneity of the originating organ. Organoids are generally considered 3D structures, but recent studies have shown that certain 2D (also called 2.5D) culture methods, such as on-a-chip approaches or 2D transwell systems^{34,35}, recapitulate the cell-type diversity of organoids. Although the definition of organoids has evolved alongside with the research, two aspects have remained constant: the first being self-organization, and the second the 3D structure^{20,36,37}. Here, we use the following definition, which encompasses structures containing epithelial derivatives but also mesenchymal components (Box 1): “An organoid is a self-organizing 3D cellular structure that can be derived from PSCs or tissue-derived stem/progenitor cells from healthy or diseased tissue. It contains key cell types and recapitulates at least some of the functions of the organ of origin.”

In this Review, we discuss recent developments in generating human organoids, and their use for studying human disease and potential clinical applications. For in-depth reviews about PSC-derived organoids, the reader is referred to other excellent reviews^{38,39}. We begin by introducing limitations, including increasing cellular and morphological complexity, and challenges of current organoid models, such as mimicking the cellular and non-cellular microenvironment of native tissues. We then highlight recent advances that at least partially overcome these limitations. The main focus is on human TSC (hTSC)-derived organoids, with comparisons to PSC organoids when relevant. We then discuss the use of organoids in drug discovery and development, and lastly, we conclude by outlining what we think will be key factors and challenges when establishing the next generation of organoids.

Limitations of TSC-derived organoids

Organoids have been successfully applied to address a wide range of questions in development, physiology and disease by complementing and in some cases having advantages over traditional models. Notable examples include modelling of human-specific cerebral cortical layers^{3,27} and the preservation of tumour cell heterogeneity that is critical for faithfully recapitulating cancer biology⁴⁰. Despite these advances, key challenges in cellular maturation, morphological complexity and clinical translation remain to be addressed before the full potential of organoid technology can be realized.

Cellular and morphological complexity

Adaptions of the culture medium have enabled the establishment of organoids from a large number of organs (Supplementary Table 1). Organoid models have greater cellular heterogeneity than 2D cell lines, but they often fail to fully recapitulate the organ complexity, owing to the lack of a specific cell type or the absence of entire cell lineages^{30,41}. For example, in expansion conditions that promote stem cell maintenance, mouse TSC-derived intestinal organoids

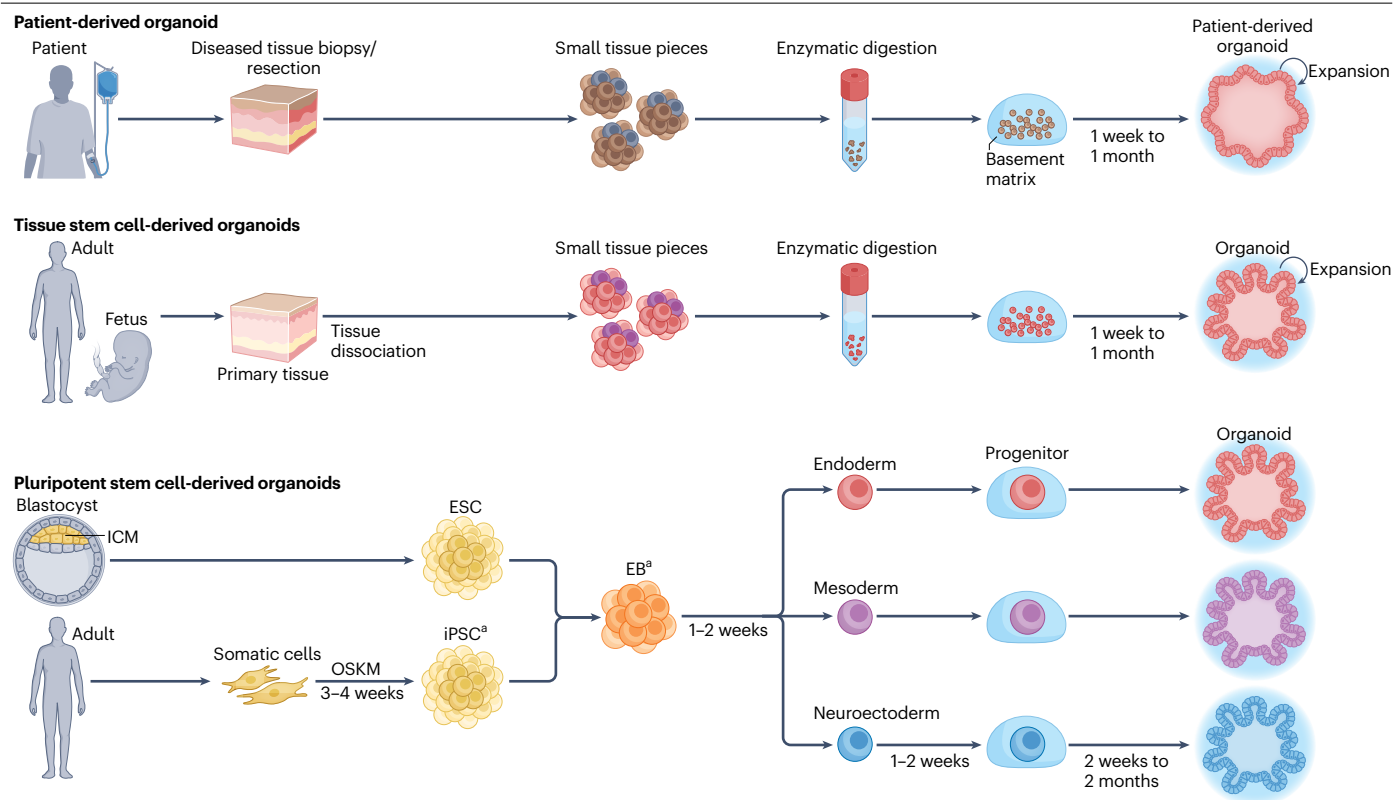


Fig. 1 | Organoid establishment. Organoids can be established from tissue stem cells (TSCs) present in primary tissues, or from pluripotent stem cells (PSCs). To establish TSC-derived organoids, primary tissue is dissociated into single cells and small fragments through a combination of mechanical means and enzymatic digestion. The resulting single cells and small cell clusters are seeded in a basement matrix. After 1 week to 1 month, the organoids reach their full size. PSC-derived organoids can be generated from human embryonic stem cells (hESCs) or induced pluripotent stem cells (iPSCs). Following reprogramming

with the Yamanaka factors (OSKM), iPSCs can be aggregated to form embryoid bodies (EB) that proceed to differentiate to contain cells of one of the three germ layers. Following differentiation of one germ layer, progenitors of the desired organ are generated and embedded in an extracellular matrix, which enable expansion and self-organization into a 3D structure, which is the organoid. In some protocols for PSC-derived organoid generation, the aggregation step takes place following germ layer specification (as indicated by 'a' in superscript).

lack of cells²⁴. Most pancreatic organoids⁴² consist solely of ductal cells, lacking acinar and endocrine cells. Similarly, hepatocytes are missing in cholangiocyte-derived organoids; PSC-derived organoids of the mesoderm and neuroectoderm usually display higher cellular heterogeneity than TSC-derived organoids, which are purely epithelial. Among the most prominent examples of PSC-derived organoids are brain organoids²⁷ and cardioids³², the latter showing co-development of cardiac epithelial and fibroblast-like cells³².

Maturation and developmental stage

The ability of organoids to recapitulate their tissue of origin depends not only on having the appropriate degree of cellular heterogeneity but also on matching the developmental stage at which the biological process of interest occurs.

PSC organoids mature to mimic organs during early to mid-gestation⁴³, and the generation of more mature cell types has proven challenging. For example, beta cells derived from PSCs can secrete insulin *in vitro*, but they do not fully match primary human beta cells in terms of insulin expression levels^{44–46}. Throughout human development, organs take months, sometimes years, to mature,

whereas *in vitro*, the differentiation time of organoids is typically only weeks to months. PSC organoids are generated through directed differentiation; however, their large size prevents nutrient access and results in a necrotic core that prevents long-term culture^{47,48}. Similarly, TSC organoids from fetal tissue remain 'locked' in a fetal state^{31–33}. Although TSC organoids can be cultured long-term, they require frequent splitting and removal of accumulated dead cells. This induces a regenerative response that promotes proliferation rather than differentiation. Spinning or rotating wall vessels^{27,47} and microfluidic chips^{49–51} have substantially extended the duration PSC and TSC organoid cultures. For the latter, there is no need for splitting (Box 2).

Translational challenges

Patient-derived organoids (PDOs) have emerged as valuable tools for translational science, bridging the gap between basic research and clinical applications. The success rate of establishing new PDOs is generally higher than that of conventional cell lines^{40,52}. However, the success rate varies depending on the organ of origin and/or tumour stage^{40,53}. For example, the success rate is similar when establishing organoids from healthy human intestinal tissue and colon cancer⁴⁰, whereas

Box 1 | Defining the term organoid

As organoids generated from primary tissue can be derived from not only adult but also embryonic and fetal stem cells, we have chosen to use the term tissue stem cell-derived organoids rather than adult stem-cell derived organoids. We want to clarify that although some organ(oid)-on-a-chip (OoC) approaches restrict organoid growth in 3D, thus becoming 2.5D, they permit overall 3D structure, and we therefore still count them as organoids. Thus, the culture environments can range from static well plates to transwells and microfluidic devices. However, OoC approaches and growth on transwells that enforce a 2D conformation no longer abides to the definition of an organoid. We refer to these as 2D organoid sheets. An alternative would have been to remove '3D' from the definition of an organoid and state that they would merely need to be able to self-organize when cultured in a 3D matrix. However, since the initial definition, the aim of an organoid has been to be organ-like, and therefore, as the organs are 3D, we see this as an important feature and have chosen to keep it in the definition. The ability to self-organize is a key feature that distinguishes organoids from spheroids. Although blastoids^{271–273}, gastruloids^{274,275}, somitoids and segmentoids^{276,277} are excellent models of developmental processes, they typically require an aggregation step before initiation of self-organization and do not recapitulate an organ but rather a process. Thus, they are by definition not organoids, although often referred to as stem cell-based embryo models or embryonic organoids.

Of note, we want to mention that adjusting the definition does not mean that the previous definitions were wrong. It only shows that in the light of new knowledge, scientific terms sometimes need to be redefined to include new advancements and developments. Although this can be seen as semantics, it is of importance to avoid misunderstandings and facilitate easy communication within the scientific community.

early-stage prostate tumours are notoriously difficult to culture⁵⁴ compared with their healthy counterpart. This highlights a need for further optimization and standardization of PDO establishment protocols.

Another key challenge is balancing the need for standardized protocols with the inherent variability of human tissues. Unlike inbred animal models, human organoids reflect the genetic diversity of their donors. This diversity can manifest as differences in cell-type composition, growth and differentiation potential, and even the efficiency of genome engineering. Additionally, organoids exhibit both inter-organoid and intra-organoid variability, such as differences in size and morphology; for example, intestinal organoids can vary in overall size and in the number of buds²⁴. Ensuring consistency will be essential for unlocking the full potential of organoids in drug screening and regenerative medicine applications. Donor variability also represents a unique strength of PDOs as it enables investigation of the influence of age, gender or genetic background and supports both personalized and precision medicine approaches. Tumour-derived PDOs recapitulate patient-specific tumour heterogeneity, especially at early passages, although genetic changes have been observed during extended culture^{53,55,56}. Finally, the use of organoids in clinical practice will require that the entire workflow complies with Good Manufacturing Practice (GMP) standards. A major obstacle is the dependence on

animal-derived ECMs⁵⁷ (such as Matrigel and BME) for long-term culture, underscoring the urgent need for suitable synthetic alternatives.

Increasing complexity

Tissue microenvironment has a crucial role in directing organ development and influencing disease progression^{58,59}. To recapitulate these conditions *in vitro*, organoids must incorporate both epithelial and non-epithelial cell types. Recent efforts have focused on integrating non-epithelial components to create more physiologically relevant models. Here, we provide examples and discuss strategies to increase the complexity of both the epithelial compartment and the non-cellular microenvironment (Fig. 2).

Increasing epithelial cellular complexity

Given the central role of the epithelium in tissue function and its frequent involvement in disease, recapitulating epithelial cellular diversity *in vitro* is essential for both fundamental mechanistic studies and translational applications.

Increasing cellular complexity using genome engineering. Developmental studies have identified key transcription factors that direct cell differentiation. By genetically modifying organoids to overexpress these factors, cell types that are not typically present in conventional cultures can be generated. For example, overexpression of the transcription factor PROX1 in human PSC (hPSC)-derived organoids promotes vascularization whereas transcriptional activation of CYP3A4, the primary drug-metabolizing enzyme of the body, increases its enzymatic activity and improves liver functions, such as FXR signalling⁶⁰. Transient overexpression of NEUROG3 in adult hTSC-derived intestinal organoids induces the differentiation of enteroendocrine cells (EECs)⁶¹. Similarly, the combined expression of Ngn3, Pdx1 and MafA has been shown to promote the generation of insulin-producing beta cells in both mouse⁶² and human⁶³ adult pancreatic ductal organoids (Fig. 2). However, forced transcription factor expression might produce 'off-target' cell types that are not found *in vivo* in native organs, highlighting the need for careful validation and comparative studies.

Increasing cellular complexity by recapitulating niche signals.

The growth medium for hTSC organoids promotes the continuous proliferation of stem and progenitor cells. However, the growth factors and signalling cues required to differentiate these cells into the various intestinal lineages can differ substantially. In the intestine, WNT and bone morphogenetic protein (BMP) signalling influence cell-fate specification along the crypt–villus axis. High WNT activity at the crypt base maintains the stem cell pool, whereas increasing BMP levels towards the villus tip promotes differentiation⁶⁴. The original culture conditions gave rise to hTSC-derived intestinal organoids that lacked two important cell types: Paneth cells, which provide antimicrobial defence, and EECs, which secrete hormones for metabolic regulation⁶⁵. Given that dysfunction of Paneth cells and EECs contributes to diseases such as inflammatory bowel disease (IBD) and metabolic disorders⁶⁶, substantial efforts have been directed towards improving differentiation protocols^{67,68} to study these disorders *in vitro*. An improved medium containing IGF and FGF2 that supports the formation of Paneth cell progenitors and EECs was developed⁶⁹. Subsequent studies have indicated that Paneth cell number and maturation can be increased by supplementing XPO1 (ref. 70), interleukin-22 (IL-22)⁷¹ or a FOXO inhibitor⁷², although at the expense of EECs. Initial differentiation strategies successfully generated endocrine cells^{69,73}, but these were

largely restricted to crypt-associated subtypes^{69,73,74}. Incorporation of BMP into the medium promoted the differentiation of villus-associated EEC subtypes⁷⁵. Beyond the intestine, recapitulating niche signals has successfully supported cell differentiation in organoids derived from the endometrium (which undergoes both glandular differentiation and luminal differentiation after hormonal stimulation)^{76,77}, lung^{78,79} and skin⁸⁰. Together, these studies illustrate how mimicking endogenous niche signals can enhance the cellular complexity and physiological relevance of epithelial organoids. Although intestinal organoids most closely approach the cellular complexity of the native tissue, many other organoids display limited cellular diversity. For example, liver organoids fail to simultaneously generate hepatocytes⁸¹ and cholangiocytes²⁹, and fetal pancreatic organoids that contain the three major cell lineages and multiple endocrine cell types have only recently been established⁸² (Fig. 2). Thus, for most organoid systems, considerable work is still required to fully recapitulate the complexity of the corresponding native organ (Supplementary Table 1). The generation of assembloids, which are self-organizing 3D systems formed by combining multiple organoids or integrating organoids with specialized missing cell types^{83,84}, is a promising approach to increase cellular complexity, particularly in tissues wherein the adult stem/progenitor cells are unipotent or the differentiation protocol gives rise to a specific compartment of the organ. Prominent examples are the generation of the ascending neural pathway by assembly of four types of regionalized PSC-derived neural organoids⁸⁵ and TSC-derived periportal liver assembloids generated by combining hepatocytes, cholangiocytes and portal mesenchymal cells⁸⁶. Notably, multi-organ assembloids can also be generated through the fusion of organoids representing different organs, with the hepato–biliary–pancreatic organ system being one of the first examples⁸⁷.

Improving organoid architecture to more closely mimic native tissues. Organoid complexity can also be increased by modifying their morphology. In some cases, optimizing the culture medium induces morphological changes. For example, mouse intestinal organoids can transition from budding to cystic morphology in the presence of high WNT concentrations⁷¹. In cystic organoids, the apical surface faces the internal lumen; however, ‘inside-out’ organoids (with the apical surface facing the exterior) have also been generated^{88–92}. To obtain higher-order structures, shape-guided strategies aim to replicate the physical boundaries present during development, thereby directing cell-type patterning and sorting⁹³. These approaches involve expanding organoids in 3D and then applying microfluidic systems or bioprinting to achieve precise spatial organization.

In an early study, collagen hydrogel was shaped to resemble intestinal crypts and villi and then placed in a modified transwell system to create growth factor gradients. Human intestinal organoids seeded onto this hydrogel displayed the expected spatial organization of stem cells and differentiated cells⁹⁴. Inspired by organ-on-a-chip (OoC) technologies, the concept of organoid-on-a-chip emerged. Combining human intestinal organoids with microfluidics has enabled the generation of villus-like⁹⁵ and crypt-like⁹⁶ structures. Continuous perfusion removes waste and dead cells, extending culture for up to ~1 month without the need for splitting. This prolonged culture period supports differentiation of rare cell types, such as tuft, enteroendocrine and early M cells, without the use of specialized differentiation media^{96,97}. The most advanced ‘intestine-on-a-chip’ models not only reproduce native tissue architecture, including crypts and villi, but also recapitulate physiological *in vivo* processes such as cell shedding at the villus

tip⁴⁹ (Fig. 2). Patient-derived adult lung⁵⁰ and pancreas⁵¹ organoids have likewise been cultured on microfluidic chips, for drug-screening applications rather than to increase morphological complexity.

Bioprinting provides an alternative engineering strategy. Instead of relying on pre-formed structures, bioinks comprising cells or organoids encapsulated within a biomaterial are printed in a layer-by-layer manner, enabling precise control over the final 3D structure^{95,98}. Added morphological complexity (relative to standard 3D culture) has been demonstrated, for example, when bioprinted intestinal organoids formed tubular structures containing self-organizing crypts⁹⁹. Another example used bioprinting to obtain assembloids containing hPSC-derived neural organoids and patient-derived glioma organoids¹⁰⁰. Beyond enhancing architectural complexity, bioprinting can increase throughput and standardize organoid production for drug screening, as shown in uniform populations of bladder cancer assembloids¹⁰¹ and PSC-derived kidney organoids¹⁰². Although organoid bioprinting remains in its infancy, these studies provide proof-of-concept that shape-guided organization can increase structural complexity. Moreover, this approach enables integration of physical cues, as discussed below (Fig. 2).

Increasing non-epithelial cellular complexity

The cellular and physical components of the microenvironment strongly influence cell behaviour and fate^{58,59}. Here, we discuss approaches to mimic such elements of the tissue microenvironment.

Replicating the physical properties of the tissue microenvironment. Replicating the complex microenvironment of native tissues *in vitro* remains a major challenge for organoid culture. Conventional

Box 2 | Studying human development using organoids

Organoids enable studies of human development by allowing access to developmental time periods inaccessible *in vivo*. PSC-derived organoids are particularly suited for studying early embryonic development.

PSC-derived organoids have also been used to study developmental disorders. Human brain organoids with mutations in CDK5RAP2 recapitulates microcephaly by being smaller in size (compared with wild-type organoids) and with a lower number of progenitors undergoing premature neurogenesis²⁷. Brain organoids have also been used to study the effect of ZIKV pathogenesis^{278–280}, and it was determined that the NS2A protein of ZIKV disrupts cortical neurogenesis²⁸⁰. Multi-chamber cardioids have been used to investigate mutational compartment-specific defects²⁸¹. Thus far, the majority of developmentally focused studies have been performed using PSC-derived organoids. However, with the recent derivation of TSC-derived fetal organoids from the lung²⁸⁴, intestine²⁸⁵ and pancreas²⁸² primary tissue and amniotic fluid²⁸³, we foresee an increase of studies focusing on human development using TSC-derived fetal organoids. A recent study by the Rawlins group has generated a single-cell atlas of the human fetal lung. They then used it to predict neuroendocrine lineage-defining transcription factors and tested these using human fetal lung organoids²⁸⁴. Together, these studies showcase how organoids can provide insights into human development.

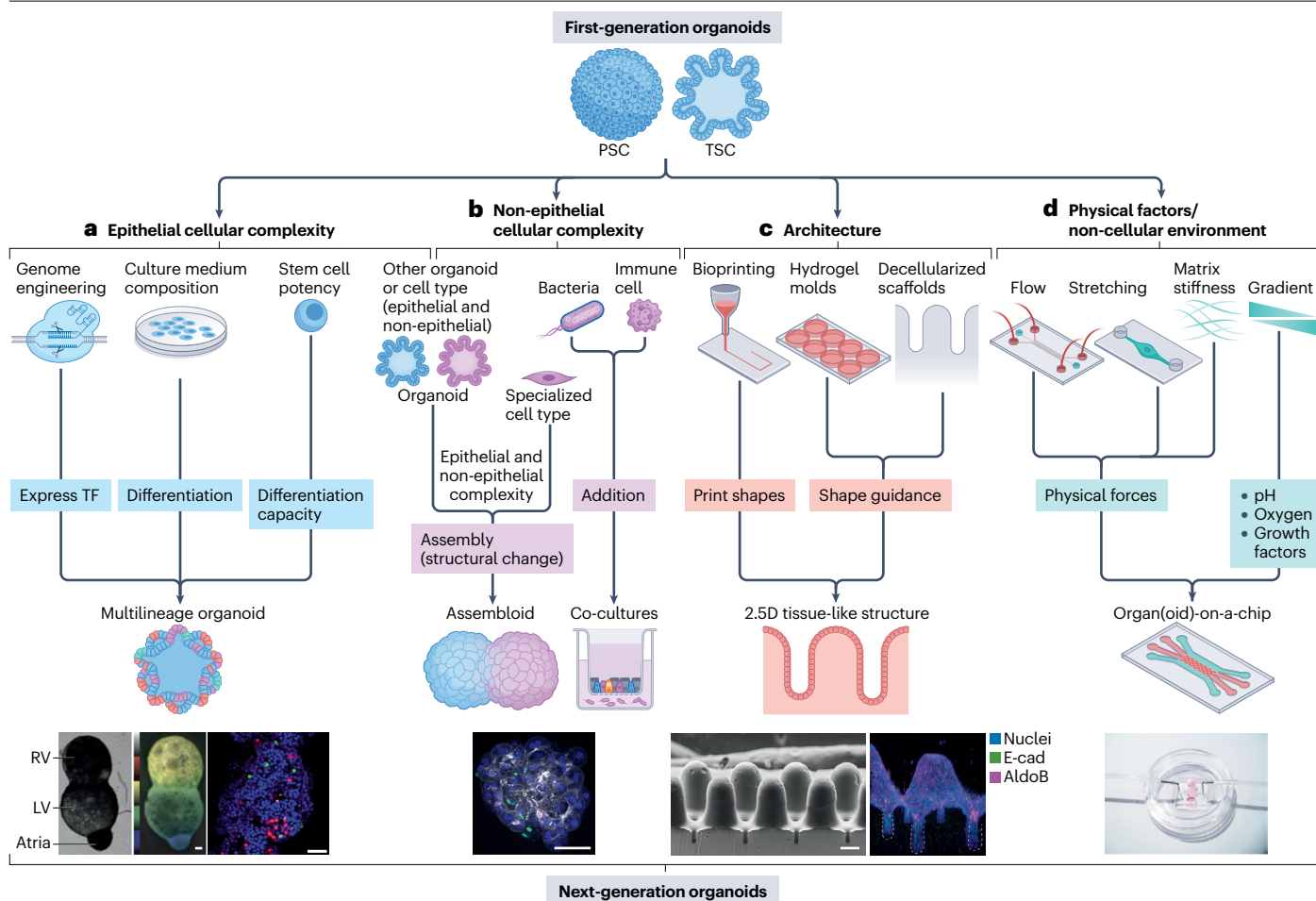


Fig. 2 | Strategies for increasing organoid complexity. Starting from first-generation organoids derived from pluripotent stem cells (PSCs) or tissue-specific stem cells (TSCs), complexity can be increased through four major strategies: epithelial cellular complexity (part **a**), non-epithelial cellular complexity (part **b**), structural (architecture) complexity (part **c**), and physical and non-cellular environmental cues (part **d**). **a**, Epithelial cellular complexity is achieved through genome engineering, culture medium modification or stem cell potency manipulation to generate multilineage organoids, two examples of which are shown (bottom). The images represent PSC-derived cardioids mimicking the atria, left ventricle (LV) and right ventricle (RV) (left and middle panels; scale bar, 200 μ m) and TSC-derived human fetal pancreatic organoid containing CHGA⁺ (red), INS⁺ (green) and SST⁺ (magenta) cells (right panel; scale bar, 50 μ m). **b**, Non-epithelial cellular complexity, introduced by adding non-epithelial cell bacteria or immune cells to organoids to form co-cultures (can be within the same extracellular matrix or in a transwell format). Both epithelial and non-epithelial complexity can be increased through assembly of multiple organoid types or specific cell types into assembloids. Image represents mouse liver assembloid. PDGFR α -H2B-GFP identifies nuclei of

portal fibroblasts (green), nuclear-tdTom highlights ductal cell nuclei (magenta), and hepatocyte nuclei are visualized in blue. Cellular architecture is visualized by phalloidin staining (white). Scale bar, 50 μ m. **c**, Structural complexity can be generated through bioprinting, hydrogel moulds or decellularized scaffolds to produce 2.5D tissue-like structures. Images (bottom) represent a custom-generated microfluidic chips for shape guidance (left panel) and TSC-derived human adult intestinal organoids cultured on a microfluidic chip with a laser-sculpted tube-shaped scaffold (right panel). The cells are stained for E-cadherin (E-cad) and aldolase B (AldoB). Scale bar, 200 μ m. **d**, Physical and non-cellular environmental cues, delivered via fluid flow, mechanical stretching, matrix stiffness or soluble gradients within organ(oid)-on-a-chip platforms. The image shows an example of a microfluidic chip. Together, these strategies yield next-generation organoids of greater physiological fidelity. Part **a** (left and middle panels) images are courtesy of Tobias Illmer, Mendjan laboratory; part **a** (right panel) image is courtesy of Amanda Andersson-Rolf. Part **b** image is courtesy of Aleksandra Slijkic, Huch laboratory. Part **c** reprinted with permission from ref. 49, AAAS. Part **d** reprinted with permission from ref. 96, Springer Nature.

submerged cultures lack the dynamic signalling gradients and physical cues present in vivo, such as the crypt-villus signalling gradients of the intestine. One strategy to increase microenvironment complexity is micropatterning¹⁰³. This approach uses precise spatial distribution of molecules on the surface of various substrates (see other reviews^{104,105}) to guide organ morphogenesis. Such biochemical patterning has

demonstrated how geometry influences neural tube morphogenesis¹⁰⁶ and crypt formation and patterning in intestinal organoids⁴⁹. Physical cues similarly shape organ function^{107,108}. For example, modulating the stiffness of synthetic matrices enables investigation of intestinal stem cell expansion⁹⁷ and liver organoid formation¹⁰⁹. Many organs, including the bladder, stomach, heart and lung, experience cyclic

stretching. Mechanical strain promotes differentiation and maturation of hPSC-derived cardiomyocytes^{110,111}, neural tube development¹¹² and intestinal crypt formation¹¹³. Elastic membranes^{110–112} or hydrogel moulds subjected to cyclic stretching¹¹⁴ can mimic physiological movements such as lung expansion during breathing^{115–117} or bladder filling¹¹⁸. Air–liquid interface transwell culture models the native air–epithelium boundary and facilitates differentiation of alveolar cells in human lung organoids^{119,120}.

To integrate chemical and physical cues in a spatiotemporally controlled manner, researchers are exploring photoresponsive ECMs and microfluidic platforms. Photoresponsive ECMs allow precise modulation of stiffness and enable controlled spatiotemporal release of bioactive substrates^{121,122}. Similarly, microfluidic systems provide control over fluid flow, pH and the formation of signalling gradients. Luminal flow has been used to mimic peristaltic movements¹²³ in hPSC-derived gastric organoids and to generate hPSC-derived pancreatic islet organoids with enhanced glucose sensitivity¹²⁴. Microfluidics has also been used to model human epiblast and amnion development from hPSCs by establishing a BMP4 gradient, illustrating the importance of this morphogen¹²⁵. For further background on OoC models, see Fig. 2 in ref. 126.

Reconstituting the physiological cellular microenvironment. Given that TSC-derived organoid models typically consist exclusively of epithelial cells, incorporating non-epithelial cell types provides a more physiologically relevant representation of the native organ microenvironment.

Co-culture approaches with hPSC-derived organoids have enabled vascularization of brain¹²⁷, liver¹²⁸ and kidney¹²⁹ organoids, as well as branching morphogenesis in fibroblast–mammary organoid co-cultures¹³⁰. To study human intestinal–immune crosstalk during development, hPSC-derived intestinal organoid models containing innate lymphoid cells (ILCs)¹³¹ and macrophages^{132,133} have been established. In these systems, the epithelial niche alone was sufficient to drive ILC maturation, whereas macrophages enhanced epithelial barrier function¹³². Human intestinal organoids can also be transplanted into humanized mice, allowing natural migration of human immune cells into the organoids. However, this approach provides less experimental control and monitoring capability than *in vitro* co-culture systems¹³⁴. Additional examples include endothelial cell-mediated support of bronchioalveolar stem-cell differentiation¹³⁵ and the recapitulation of liver portal tract architecture in ductal cell–liver mesenchymal cell co-cultures¹³⁶. Immune–epithelial interactions have likewise been studied in fetal TSC-derived intestinal organoids¹³⁷. The first adult intestinal immuno-organoid model integrates autologous tissue-resident memory T cells into human intestinal organoids to investigate mechanisms of drug-induced intestinal inflammation¹³⁸ (Fig. 2).

Disease modelling

Organoids offer distinct advantages as preclinical models for translational research. Organoids derived from healthy tissue offer a unique advantage for disease modelling by enabling a reductionist, bottom-up approach using isogenic controls. Genome engineering can be applied to directly assess the functional consequences of genetic variation. In this section, we highlight examples in which genome editing of hTSC-derived organoids has provided mechanistic insights into human disease. For in-depth reviews of CRISPR–Cas-based approaches, we refer the reader to refs. 139–141.

Modelling disease using genome engineering

The combination of genome engineering and organoids has opened new possibilities to model human disease with greater and/or enhanced precision and physiological relevance. Beyond targeted applications, such as introducing defined genetic alterations or reconstructing tumorigenic mutation sequences, CRISPR–Cas9-based approaches also enable unbiased exploration of novel regulators of development and disease.

Modelling non-neoplastic disorders. Monogenic diseases, arising from a single genetic alteration, are relatively straightforward to model using genome engineering techniques. For example, CRISPR–Cas9-mediated disruption of the enzyme DGAT1 (diacylglycerol *O*-acyltransferase 1), which catalyses the rate-limiting step of triacylglycerol formation in intestinal organoids, has been used to model intractable diarrhoea and malabsorption¹⁴². Likewise, fetal hepatocyte organoids with engineered mutations in genes associated with lipid disorders have been used to model steatosis – the first stage of non-alcoholic fatty liver disease – and to screen for drugs that resolve this condition¹⁴³. Gene editing has also been applied to correct CFTR mutations in cystic fibrosis PDOs^{144,145}, and it has been used to model the disease¹⁴⁶. Both base and prime editing has also been used on PSC organoids^{147,148}. Finally, kidney tubuloids have been used to model autosomal dominant polycystic kidney disease¹⁴⁹. Together, these studies, although still limited in number, illustrate the potential of combining organoid technology with genome editing to model non-neoplastic diseases.

Genetic reconstruction of human tumorigenesis. The number of studies focused on modelling tumorigenesis by far outnumbers those on non-neoplastic disorders. Besides highlighting milestones, we also discuss recent work focused on modelling tumorigenesis of the intestine and pancreas. Among the earliest applications of CRISPR–Cas9 technology to introduce targeted mutations in hTSC-derived organoids was the generation of a frameshift mutation in adenomatous polyposis coli (APC), a negative regulator of the WNT signalling pathway. The resulting organoids acquired the ability to grow independently of exogenous WNT signals¹⁴⁴. This strategy of positive selection by removal of specific growth factors or addition of small molecule inhibitors was further used to introduce sequential mutations to model both microsatellite-stable^{150,151} and microsatellite-unstable¹⁵² colorectal cancer (CRC) tumorigenesis.

Beyond APC, other components of the WNT signalling pathway have been investigated using genome editing in hTSC-derived colon organoids to elucidate their roles in CRC development. For example, truncating mutations in the E3 ubiquitin ligase RNF43, which targets the Frizzled receptor for degradation¹⁵³, recapitulate early-onset CRC in organoid models¹⁵⁴. Furthermore, the combined loss of RNF43 and TP53 results in niche-independent organoid growth. Organoids have also been used to investigate the functional consequences of less well-characterized genetic alterations in CRC, including gene knockouts¹⁵⁵ and gene fusions¹⁵⁶. The ability to recapitulate tumorigenesis *in vitro* is particularly valuable for studying rare cancers with limited patient samples. This approach has recently been used to model colorectal neuroendocrine tumours and carcinomas¹⁵⁷. Furthermore, a recent study generated multi-mutant colon organoids in a single step using CRISPR base editor multiplexing¹⁵⁸. This level of precision is crucial, as different point mutations within the same gene can have distinct functional consequences, as exemplified for TP53 (ref. 13) and KRAS¹⁵⁹.

Transplantation of genome-engineered intestine or colon organoids showed that the number of mutations reflects not only tumorigenic

Table 1 | Genome engineering in organoids

Type of genome engineering	Type of organoid	Application	Refs.
Viral overexpression	PSC	Disease modelling	271
	TSC	Method or protocol Reporter EEC differentiation Knockdown	61,272–275
Transposase overexpression	PSC	Optogenetics	276
	TSC	Method or protocol Disease modelling Gene function	151,165,277
Knockouts	PSC	Conditional knockout Screens	278,279
	TSC	Disease modelling Screens	150–152, 163,164
Knock-ins (reporters)	PSC	Cell-type reporters	280
	TSC	Method or protocols Tuft cell reporter Gene correction	82,144,281
Base editing	PSC	Gene correction	147
	TSC	Gene correction	145,158
Prime editing	PSC	Gene correction	148
	TSC	Gene correction Disease modelling	282,283

EEC, enteroendocrine cell; PSC, pluripotent stem cell; TSC, tissue stem cell.

capacity¹⁵¹ but also the ability of the mutant organoids to metastasize¹⁶⁰. The observation that increasing mutational burden enhances tumorigenicity was also reported for liver cholangiocyte¹⁶¹ and ductal pancreatic organoids¹⁶². By introducing mutations in KRAS and TP53, characteristics of pancreatic ductal adenocarcinoma (PDAC) were observed after transplantation. A contrasting study targeting the same genes, observed PanIN, a precursor stage, following transplantation. This discrepancy could be due to overexpression rather than knock-in of the KRAS^{G12V} mutation or differences in transplantation site (subcutaneous vs orthotopic). Beyond these examples, tumour-suppressor and oncogenic mutations have been introduced in a wide range of organoid systems (Table 1).

Genetic screens to identify novel regulators. The focus of genome engineering has been to investigate the functional consequences of mutations identified through next-generation sequencing. Genetic screens offer a complementary approach to uncover novel regulators of developmental and disease processes. In cancer research, targeted¹⁶³ and genome-wide^{164,165} genetic screens in colon organoids have identified tumour suppressor genes mediating TGF- β resistance^{163,164}. In the context of studying development, CRISPR screens have uncovered key factors involved in progenitor cell maintenance, such as SMARCA4/SMARCC1 (components of the SWI/SNF (BAF) chromatin remodelling complex) in the fetal intestine¹⁶⁶, and SOX9 (a transcription factor with an important role in stem/progenitor cell maintenance and early cell fate) in the fetal lung¹⁶⁷. The zinc finger protein ZNF800, which acts as a transcriptional repressor, was also identified as a novel regulator

of intestinal EECs¹⁶⁵. These studies highlight how the combination of genome engineering and organoid technologies can lead to the identification of novel regulators of developmental and disease processes.

Modelling disease using co-culture models

Modulating the microenvironment through co-culture techniques can further enhance the physiological relevance of disease models. Co-culture approaches have focused on incorporating non-epithelial cells into PDOs.

Modelling the tumour microenvironment. Neoplastic transformation of healthy epithelial cells initiates a dynamic interplay with the surrounding tissue. This interaction induces molecular, cellular and physical changes in adjacent tissue-resident cell types, including fibroblasts, immune cells and endothelial cells, which together form the tumour microenvironment (TME)¹⁶⁸. The crosstalk between tumour cells and immune cells within the TME is bidirectional. CRC cells can modulate T cell responses by altering the activity of antigen-presenting cells. For example, metastatic CRC PDOs impair the ability of dendritic cells to induce allogeneic T cell proliferation¹⁶⁹. Furthermore, CRC cells expressing high levels of SIRT, a protein associated with CRC prognosis, can directly suppress CD8⁺ T cell proliferation and antitumour activity via tumour-associated macrophages¹⁷⁰. Similarly, epithelial PDAC cells stimulate natural killer cells to produce elevated levels of intracellular IL-10, an immunosuppressive cytokine¹⁷¹. Immune cells can also influence tumour cell behaviour: co-culture studies of PDAC and gastric cancer organoids with myeloid-derived suppressor cells show reduced sensitivity of tumour organoids to anti-PD1 therapy^{172,173}. More recent work has expanded co-culture models to include additional cell types, such as neurons¹⁷⁴ and cancer-associated fibroblasts^{175,176}.

Modelling inflammation. The most common chronic inflammatory condition of the intestine is IBD. IBD is considered a multifactorial disorder characterized by chronic intestinal inflammation with unknown aetiology^{177,178}. Cytokines profoundly affect colonic epithelial function¹⁷⁹ and can contribute to CRC carcinogenesis¹⁸⁰. Co-culture experiments with immune cells suggest that dysfunctional T cells, through their production of IFN- γ , may contribute to epithelial cell death and tissue injury in IBD¹⁸¹. Epithelial cell death was also observed in a co-culture model of mucosal T lymphocytes and epithelial cells designed to mimic aspects of Crohn's disease¹⁸². In addition to T cells, macrophages contribute to intestinal inflammation. Co-culture of hTSC-derived colon organoids with monocyte-derived macrophages, followed by lipopolysaccharide stimulation, resulted in increased production of pro-inflammatory cytokines, including CXCL10, IL-1 β and IL-6 (refs. 183,184). The role of the microbiome in IBD pathophysiology is also being investigated using organoid models. For example, *Clostridium difficile*, a common cause of antibiotic-associated diarrhoea and colitis, has been shown to inhibit the Na⁺/H⁺ exchanger 3 (NHE3) in hPSC-derived colon organoids. In vivo, loss of NHE3 in mice alters the intestinal environment, favouring *C. difficile* colonization¹⁸⁵ and potentially contributing to IBD pathogenesis¹⁸⁵. Conversely, *Lactobacillus* species exert a protective effect by promoting regeneration of the intestinal mucosa¹⁸⁶.

Studying tissue regeneration

Stem cell-based regenerative medicine offers the potential to restore damaged or lost tissues. Organoids have been used to study mechanisms of tissue regeneration and their potential for cell replacement therapy.

Studying regeneration in vitro

Immune cells also have an important role during post-injury tissue regeneration. Notably, ILs secreted by immune cells affect intestinal barrier homeostasis¹⁸⁷. IL-27 promotes regeneration by stimulating cell proliferation and enhancing the expression of claudins (transmembrane proteins and components of tight junctions) and mucous production in colon organoids previously exposed to inflammatory cytokines¹⁸⁸. The role of IL-22 remains controversial, but some studies suggest that it promotes recovery of the epithelial barrier following acute injury by stimulating proliferation of LGR5+ stem cells in mice and human intestinal organoids^{189,190}. An IL-22–IgG4 Fc (the constant fragment of the immunoglobulin heavy chain) fusion protein is currently in clinical trials for patients with moderate-to-severe ulcerative colitis and Crohn's disease¹⁹¹. Organoid models are also being used to investigate the molecular mechanisms underlying epithelial regeneration. Studies utilizing mouse TSC-derived organoids have identified YAP signalling as a central player in the regenerative response of both the intestine¹⁹² and the liver¹⁹³. In the lung, Notch signalling and the transcription factor Fosl2 mediate the conversion of secretory cells into alveolar type 2 cells following injury¹⁹⁴.

Transplantation to promote in vivo regeneration

Cell replacement therapy using ex vivo-generated cells represents a promising approach for regenerating damaged or lost tissues. Organoids, with their capacity for self-renewal and differentiation, offer a potential source of cells for such therapies. Proof-of-concept transplantations of hTSC-derived organoids into immunodeficient mice have been carried out for multiple organoid types, including colon¹⁹⁵, conjunctiva^{196,197}, liver²⁹ and pancreas^{198,199}. Transplanted PSC-derived and fetal TSC-derived organoids can lead to terminal cell maturation that may not occur in vitro²⁰⁰. This approach has been used to ameliorate short bowel syndrome (SBS), a condition characterized by malabsorption and wasting that arises from extensive intestinal resection. In a rat model of SBS, replacing the native colonic epithelium with ileum-derived organoids reduced intestinal failure. This beneficial effect was attributed to the nutrient absorptive function of the transplanted intestinal organoids, as transplantation of colon organoids did not improve outcomes²⁰¹. Transplantation of larger numbers of cells necessitates scaling up the organoid generation procedure. Cellular bioprinting of kidney tissue sheets with functional proximal tubular segments¹⁰² and 3D centimetre-scale tubular intestinal epithelia⁹⁹ has demonstrated how engineering solutions can generate larger tissue-like structures.

Translating organoid research into clinical applications has notable challenges, including the need to adapt laboratory protocols to

meet GMP standards²⁰². Despite these hurdles, clinical trials evaluating the safety and efficacy of hPSC-derived islet-like organoids for the treatment of diabetes are ongoing^{203,204}. The first clinical trial using hTSC-derived organoids was initiated recently and involved the autologous transplantation of salivary gland organoids to treat radiation-induced hyposalivation and xerostomia (dry mouth syndrome)²⁰⁵. This translation of organoid technology marks an important milestone in the potential use of stem cells in regenerative medicine.

Organoids in drug discovery and development

The high failure rate of drug candidates in clinical trials is in part due to limitations of traditional preclinical models²⁰⁶. These models do not capture the pathophysiology and heterogeneity of human tissues, leading to inaccurate predictions of drug response and toxicity. By preserving individual patient variability, organoids offer a platform for more physiologically relevant drug screen and toxicity assessments and more accurate predictions, and may thus increase the success rates of clinical drug development (Fig. 3).

Drug development and discovery screens

Organoid biobanks are valuable resources for investigating disease heterogeneity and facilitating precision medicine approaches. Although organoids are useful for basic research, such as disease modelling and drug target identification, their potential extends to later stages of the drug development process. Here, we highlight the use of organoids in drug target identification and compound screening, as well as in preclinical efficacy and toxicity assessments. As gene therapy has been extensively discussed in multiple recent reviews, we refer the reader to these^{140,207}. Initial drug screens using PDOs typically involved a limited number of organoid lines and fewer than 100 compounds. These screens were conducted manually, which restricted throughput and scalability^{40,162}. Recent advances, including the establishment of large-scale biobanks with over 100 organoid lines^{145,208}, and technological innovations are now overcoming these limitations^{209,210}. The use of patient-matched isogenic organoid lines enables direct assessment of tumour-specific drug efficacy. This approach has been applied to test both single-agent^{211,212} and combinatorial therapies²¹³ across multiple cancer types^{157,214,215}. The ability to generate and cryopreserve PDOs of rare cancer types enables the collection of larger cohorts, facilitating growth factor dependency studies and drug screening efforts^{157,214,215}. Drug repurposing provides a cost-effective and efficient strategy for identifying new therapeutic applications for existing drugs. Organoid biobanks are proving valuable for such screens, as exemplified by the high-throughput screen of 1,172 FDA-approved compounds on PDAC PDOs, which identified 26 effective compounds, including 19 already

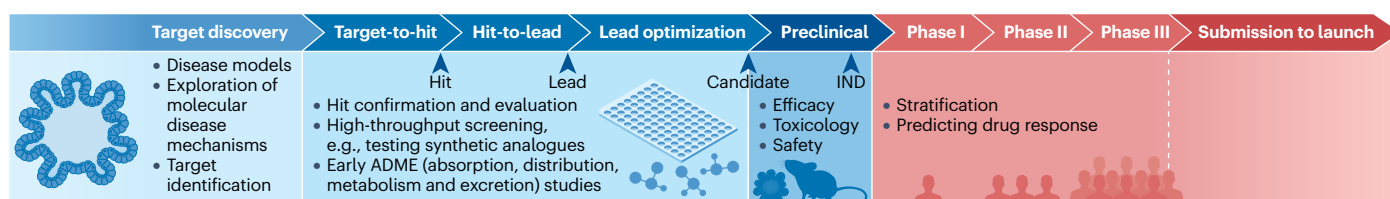


Fig. 3 | Applications of organoid technology across the drug discovery and development pipeline. The drug discovery and development pipeline comprises several sequential phases: target identification and validation, hit and lead discovery, lead optimization, preclinical testing and clinical trials (phases I–III), culminating in regulatory approval and post-approval monitoring. Organoid-

based models can contribute meaningfully at each stage by serving as disease models during target validation, as screening platforms for hit identification, as toxicity and efficacy readouts during lead optimization and preclinical testing, and finally, as patient-derived tools for predicting clinical responses and supporting precision medicine approaches. IND, Investigational New Drug.

approved for other cancer types²¹⁶. Similar repurposing efforts have been conducted using primary²¹⁷ and metastatic²¹⁸ CRC PDOs, as well as for non-neoplastic diseases such as cystic fibrosis²¹⁹ and IBD^{220–222}.

Biomarkers have a crucial role in guiding patient treatment strategies, helping mitigate toxic side effects and reduce treatment costs. Recent studies using gastric²²³, head and neck²²⁴, and colon cancer organoid²²⁵ biobanks have provided proof-of-concept for their potential to identify and validate new biomarkers. Whereas both personalized and precision medicine use biomarkers to inform treatment decisions, functional precision medicine relies on functional testing. By performing ex vivo perturbations on tumour cells, sensitivity and resistance data can guide treatment strategy. Numerous studies have highlighted the predictive value of organoids in determining patient-specific drug responses across various cancer types, including colon, mesothelioma and ovarian cancer^{226–232}. However, there are exceptions. In a screen of PDOs from metastatic CRC lesions, organoids failed to predict responses to combined 5-FU and oxaliplatin²³³. This discrepancy was attributed to the absence of the in vivo microenvironment. Indeed, subsequent co-culture studies with stromal cells have indicated that stroma can confer therapy resistance^{234,235}. These studies have also revealed that PDOs are capable of instructing tumour subtype-specific stromal changes²³⁵. Ex vivo tumour fragments obtained directly from patient material^{236–238} retain the integrity of the TME, but the limited material constrains the number of drugs that can be tested, even when using microscale techniques²³⁹. Despite being derived from a single spatial location, PDOs generally show strong efficiency in recapitulating patient responses²⁴⁰. A recent study has evaluated the expression levels of target genes for sorafenib and lenvatinib (first-line treatments) in 255 liver cancer PDOs derived from multiple spatially distinct tumour regions from 79 patients. Variations in target-gene expression levels between PDOs from different regions correlated with drug resistance, with lower expression associated with increased resistance²⁴⁰.

Applications in cancer immunotherapy

Despite considerable advances in cancer immunotherapy, only a fraction of patients experiences durable clinical benefit. Moreover, many patients who initially respond to immunotherapy eventually develop resistance, either through inherent mechanisms or acquired resistance that emerges during treatment^{241,242}. Intra-patient and inter-patient tumour heterogeneity, together with variations in immune responses, further complicates immunotherapeutic approaches^{241–243}.

Organoids are increasingly being used to investigate tumour-immune interactions in disease progression and treatment resistance. Immune-organoid co-cultures can partially recapitulate the native tumour immune microenvironment, including patient-specific variation^{244,245}. Co-culture of different PDAC PDOs with T cells results in distinct T cell subset composition, reflecting patient-specific differences in immune responses²⁴⁵.

Immune-organoid co-cultures also provide a promising platform for the preclinical evaluation of chimeric antigen receptor T cell (CAR-T) therapies. Co-culturing PDOs with their endogenous tumour-infiltrating lymphocytes indicated that immune checkpoint blockade promotes the expansion and activation of tumour-specific T cells, ultimately inducing tumour cytotoxicity²⁴⁶. Proof-of-concept studies using bladder²⁴⁷ and cholangiocarcinoma²⁴⁸ PDOs likewise have demonstrated that these models can predict both efficacy and toxicity in a patient-specific manner. CAR-NK-92 cells were used in a cytotoxicity assay targeting the neoantigen EGFRvIII on CRC PDOs²⁴⁹. A novel CAR strategy was also evaluated, targeting Frizzled receptors,

which are overexpressed in colorectal tumours lacking RNF43 and ZNRF3. However, this approach resulted in on-target–off-tumour toxicity²⁴⁹. Such toxicity has also been observed with CAR-T therapies and T cell-engaging bispecific antibodies (TCBs), which redirect T cell cytotoxicity towards any cell expressing the target antigen, malignant or not. Recently, these effects were investigated using a donor-matched intestinal organoid–peripheral blood mononuclear cell co-culture model with TCBs targeting the antigens EPCAM and CEA. Addition of TCBs induced apoptosis in the organoids. In vivo, this would be detrimental for the intestinal epithelium and might explain the adverse effects seen in phase I clinical trials (Fig. 3). Importantly, this study highlights the value of human-derived model systems, as this on-target–off-tumour toxicity had not been detected in the mouse models used during preclinical development²⁵⁰.

Organoids for preclinical toxicity testing

Among the drug candidates that reach clinical trials, ~30% fail owing to unmanageable toxicity²⁵¹. This failure often stems from the inability to capture species-specific differences in drug uptake and metabolism during the preclinical phase²⁵². Key organs involved include the intestine (drug uptake), liver (metabolism) and kidney (excretion)²⁵³. Here, we discuss how organoids derived from these organs can help predict drug metabolism and toxicity. In vivo, nutrient absorption is mediated by mature intestinal enterocytes expressing various nutrient transporters. To accurately model this process, culture conditions have been optimized to promote enterocyte maturation and transporter expression in both mouse²⁵⁴ and human small intestinal organoids²⁵⁵. Most drugs are metabolized by cytochrome P450 enzymes (CYPs), which have a crucial role in drug clearance and potential drug–drug interactions. Expression of these enzymes has been successfully induced in both TSC-derived^{256,257} and PSC-derived²⁵⁸ intestinal organoids. Together with the latest addition of human intestinal immuno-organoids¹³⁸, these studies provide a starting point for investigating the complex interplay between drug uptake, metabolism and transport in a more physiologically relevant setting.

The liver metabolizes drugs into substances that can be eliminated from the body. Disruption of this process can result in drug-induced liver injury (DILI), causing unmanageable toxicity²⁵⁹. DILI is patient-specific and has been hard to detect in the preclinical phase. Although current liver hepatocyte organoid models may still not recapitulate the maturity and functionality of adult hepatocytes, they have demonstrated potential for predicting DILI in vitro^{259,260}. Drug-induced nephrotoxicity is another serious adverse effect. PSC-derived kidney organoids expressing key transporters involved in drug excretion have been developed to model drug-induced nephrotoxicity^{261–263}. These models enable the assessment of this toxicity in a segment-specific manner, providing another valuable tool for preclinical drug safety evaluation (Fig. 3).

Conclusions and perspectives

Organoids are set to continue delivering important advances in fundamental biology and biomedical applications. Progress in the field will come from balancing developing more complex models while maintaining the advantages of simpler systems to study diverse biological processes, including otherwise inaccessible human developmental mechanisms, as well as for disease modelling and drug testing.

Although the earlier organoid models did not recapitulate the complete native microenvironment, this limitation is also a strength, offering a reductionist approach to study epithelial systems in

isolation. For example, combining organoids with genome engineering enables bottom-up reconstruction of complex diseases such as cancer, providing nuanced insight into their underlying genetic complexities. This approach makes it possible to generate larger cohorts of rare-disease organoid models using genome engineering or by establishing living biobanks. Furthermore, genome engineering enables the generation of reporters and lineage tracing constructs, and we anticipate its central role in dissecting functional dynamics and molecular mechanisms to address increasingly complex biological questions.

The repertoire of organoids is expanding beyond mouse and human, and we predict that 'first-generation' models from additional species will provide new knowledge through evolutionary comparisons, species-specific physiology and, as demonstrated by a recent bat organoid collection²⁶⁴, studies of viral infection mechanisms and tropism.

Increasing organoid complexity enables investigations of interactions between different cell types and more holistic studies of human development and disease. Hence, there are clear advantages and a growing need to further increase the complexity of current models.

Although the culture medium supplies the necessary factors for differentiation of specific cell types, the outcome depends on the intrinsic potency of the organoids' stem/progenitor cell population. The starting material is, therefore, of utmost importance. For adult organs in which the stem/progenitor cells are unipotent, we envision that assembloids and OoC approaches will help increase epithelial heterogeneity. Examples of the assembly approach include the generation of liver buds from induced PSCs²⁶⁵. A recent example shows how assembloids of adult hepatocytes, cholangiocytes and mesenchymal cells can recapitulate liver periportal architecture and allow modelling of cholestatic injury and biliary fibrosis^{86,266}. Although we are moving towards more complex co-cultures containing non-epithelial cell types, much work remains before endogenous epithelial complexity can be fully recapitulated. As the assembloids approach does not require expensive specialized equipment or extensive technical training, it is relatively easily accessible, and we therefore expect this approach to be used more frequently in the future.

Co-cultures will be essential to investigate interactions between epithelial cells and their microenvironment, such as mesenchymal, neuronal and immune cells, but also to study interactions between different epithelial tissues, such as placental organoids and decidua²⁶⁷. Overall, organoid co-culture technology remains at an early stage, yet we foresee progress driven by assembloid approaches and the synergy between organoid and engineering approaches. Recently, this potential was elegantly demonstrated by combining genome engineering, optogenetics, OoC methodologies and live imaging to study colon cancer initiation over several weeks²⁶⁸.

The integration of organoid technology with artificial intelligence is another area wherein we predict strong synergy. The latest artificial intelligence models predict the structure of complexes, including proteins, nucleic acids and small molecules²⁶⁹, and peptide binding specificity²⁷⁰, and they help generate unprecedented volumes of hypotheses based on morphological and functional data. Translating these predictions into real-world clinical benefit requires rigorous testing and validation. Organoids will be essential to this process by providing platforms for functional testing and generating human *in vitro* datasets at scale, which are key for training artificial intelligence models to enhance their accuracy in predicting human physiology.

Pharma and biotech companies are already adopting organoids, and their integration into the drug discovery and development pipeline

Glossary

2D transwell systems

Permeable membrane inserts that allow polarized cell monolayers to be cultured at an air-liquid or liquid-liquid interface, enabling separate access to apical and basolateral compartments.

Assembloids

Self-organizing 3D cellular systems formed by the combination of organoids with other organoid types or with distinct lineages; notably, self-organization and structural change are defining criteria of assembloids.

Antigen

A molecule (for example, protein or polysaccharide) that is typically recognized by the adaptive immune system and capable of eliciting a specific antibody or T cell response.

Chimeric antigen receptor T cell

(CAR-T). A genetically engineered T cell expressing a synthetic receptor that directs cytotoxic activity against a specific tumour-associated antigen, independent of major histocompatibility complex presentation.

Crohn's disease

A chronic transmural inflammatory bowel disease that can affect any segment of the gastrointestinal tract, commonly presenting with skip lesions and extraintestinal manifestations.

Cytochrome P450 enzymes

(CYPs). A superfamily of haem-containing monooxygenases responsible for the phase I oxidative metabolism of most clinically used drugs, primarily expressed in the liver.

Isogenic controls

Cell lines or organoids that are genetically identical to an experimental line except for the specific mutation of interest, providing the background control for disease modelling.

M cells

Specialized epithelial cells overlying intestinal Peyer's patches that transcytose luminal antigens and microorganisms to underlying immune cells to initiate mucosal immune responses.

On-a-chip approaches

Microfluidic devices for culturing cells or organoids within miniaturized chambers engineered to recapitulate organ-level mechanical forces, fluid flow and compartmentalized tissue architecture.

Pancreatic ductal adenocarcinoma

(PDAC). One of the most lethal cancer types arising from the exocrine ductal epithelium of the pancreas; known for late diagnosis, stromal desmoplasia and poor prognosis.

Ulcerative colitis

A chronic inflammatory bowel disease characterized by relapsing mucosal inflammation confined to the colon and rectum.

is underway. An outstanding issue for clinical use of organoids is the need to adapt protocols to meet GMP guidelines. A central challenge is the dependence on animal-derived ECMs, such as Matrigel and BME. Although synthetic alternatives have been developed, none currently support long-term organoid growth. Consequently, developing a GMP-compliant ECM capable of sustaining long-term culture remains a major goal, and efforts to attain this are ongoing. Adding complexity to organoid models has many benefits, but it also introduces additional variability and, therefore, requires robust and well-defined readouts. As the field progresses, striking the right balance between complexity and simplicity becomes increasingly important. The level of complexity should align with the specific application, purpose and research

question. We therefore envision that the next generation of organoids will complement rather than replace the existing models, and together, these systems will increasingly contribute to the understanding of human development and disease.

Published online: 11 May 2026

References

- Mallo, M., Wellik, D. M. & Deschamps, J. Hox genes and regional patterning of the vertebrate body plan. *Dev. Biol.* **344**, 7–15 (2010).
- Mouse Genome Sequencing Consortium. Initial sequencing and comparative analysis of the mouse genome. *Nature* **420**, 520–562 (2002).
- Lui, J. H., Hansen, D. V. & Kriegstein, A. R. Development and evolution of the human neocortex. *Cell* **146**, 18–36 (2011).
- Guerin, M. V., Finisguerra, V., Van den Eynde, B. J., Bercovici, N. & Trautmann, A. Preclinical murine tumor models: a structural and functional perspective. *eLife* **9**, e50740 (2020).
- Inoue, T. et al. CYP2C9-catalyzed metabolism of S-warfarin to 7-hydroxywarfarin in vivo and in vitro in chimeric mice with humanized liver. *Drug Metab. Dispos.* **36**, 2429–2433 (2008).
- Sanoh, S. et al. Predictability of metabolism of ibuprofen and naproxen using chimeric mice with human hepatocytes. *Drug Metab. Dispos.* **40**, 2267–2272 (2012).
- Pan, X., Yang, Y. & Zhang, J.-R. Molecular basis of host specificity in human pathogenic bacteria. *Emerg. Microbes Infect.* **3**, e23 (2014).
- Rothenburg, S. & Brennan, G. Species-specific host–virus interactions: implications for viral host range and virulence. *Trends Microbiol.* **28**, 46–56 (2020).
- Peek, R. M. Jr & Blaser, M. J. *Helicobacter pylori* and gastrointestinal tract adenocarcinomas. *Nat. Rev. Cancer* **2**, 28–37 (2002).
- Miner, J. J. et al. Zika virus infection during pregnancy in mice causes placental damage and fetal demise. *Cell* **165**, 1081–1091 (2016).
- Gladwyn-Ng, I. et al. Stress-induced unfolded protein response contributes to Zika virus-associated microcephaly. *Nat. Neurosci.* **21**, 63–71 (2018).
- Bao, L. et al. The pathogenicity of SARS-CoV-2 in hACE2 transgenic mice. *Nature* **583**, 830–833 (2020).
- Duryee, W. R. & Doherty, J. K. Nuclear and cytoplasmic organoids in the living cell. *Ann. N. Y. Acad. Sci.* **58**, 1210–1231 (1954).
- Pomerat, C. M., Lefebvre, C. G. & Smith, M. Quantitative cine analysis of cell organoid activity. *Ann. N. Y. Acad. Sci.* **58**, 1311–1321 (1954).
- Wolter, J. R. Proliferating pigment epithelium. Producing a simple organoid structure in the subretinal space of a human eye. *Arch. Ophthalmol.* **77**, 651–654 (1967).
- Gordienko, S. M. Organoid teratoma of the nose in an infant. *Vestn. Otorinolaringol.* **26**, 92–94 (1964).
- Pinkus, H. Organoid nevus. *Mod. Probl. Paediatr.* **20**, 50–57 (1976).
- Hachitanda, Y. & Tsuneyoshi, M. Neuroblastoma with a distinct organoid pattern: a clinicopathologic, immunohistochemical, and ultrastructural study. *Hum. Pathol.* **25**, 67–72 (1994).
- Zimmermann, B. Lung organoid culture. *Differentiation* **36**, 86–109 (1987).
- Lancaster, M. A. & Knoblich, J. A. Organogenesis in a dish: modeling development and disease using organoid technologies. *Science* **345**, 1247125 (2014).
- Li, M. L. et al. Influence of a reconstituted basement membrane and its components on casein gene expression and secretion in mouse mammary epithelial cells. *Proc. Natl Acad. Sci. USA* **84**, 136–140 (1987).
- Bissell and co-workers show that Matrigel can be used to promote 3D structure formation of epithelial cells.**
- Eiraku, M. et al. Self-organized formation of polarized cortical tissues from ESCs and its active manipulation by extrinsic signals. *Cell Stem Cell* **3**, 519–532 (2008).
- Eiraku et al. describe self-organization from ESCs.**
- Eiraku, M. et al. Self-organizing optic-cup morphogenesis in three-dimensional culture. *Nature* **472**, 51–56 (2011).
- Sato, T. et al. Single Lgr5 stem cells build crypt-villus structures in vitro without a mesenchymal niche. *Nature* **459**, 262–265 (2009).
- Sato et al. describe the establishment of the first TSC-derived organoid from the mouse small intestine.**
- Ootani, A. et al. Sustained in vitro intestinal epithelial culture within a Wnt-dependent stem cell niche. *Nat. Med.* **15**, 701–706 (2009).
- Sato, T. et al. Long-term expansion of epithelial organoids from human colon, adenoma, adenocarcinoma, and Barrett's epithelium. *Gastroenterology* **141**, 1762–1772 (2011).
- Sato et al. describe the establishment of the first hTSC-derived organoid.**
- Lancaster, M. A. et al. Cerebral organoids model human brain development and microcephaly. *Nature* **501**, 373–379 (2013).
- Lancaster et al. establish the first hPSC-derived brain organoids.**
- Sampaziotis, F. et al. Cholangiocytes derived from human induced pluripotent stem cells for disease modeling and drug validation. *Nat. Biotechnol.* **33**, 845–852 (2015).
- Huch, M. et al. Long-term culture of genome-stable bipotent stem cells from adult human liver. *Cell* **160**, 299–312 (2015).
- Boj, S. F. et al. Organoid models of human and mouse ductal pancreatic cancer. *Cell* **160**, 324–338 (2015).
- Hendriks, D. et al. Human fetal brain self-organizes into long-term expanding organoids. *Cell* **187**, 712–732.e38 (2024).
- Hofbauer, P. et al. Cardioids reveal self-organizing principles of human cardiogenesis. *Cell* **184**, 3299–3317.e22 (2021).
- Takasato, M. et al. Kidney organoids from human iPS cells contain multiple lineages and model human nephrogenesis. *Nature* **526**, 564–568 (2015).
- Wright, C. W. et al. Establishment of a 96-well transwell system using primary human gut organoids to capture multiple quantitative pathway readouts. *Sci. Rep.* **13**, 16357 (2023).
- Moon, C., VanDussen, K. L., Miyoshi, H. & Stappenbeck, T. S. Development of a primary mouse intestinal epithelial cell monolayer culture system to evaluate factors that modulate IgA transcytosis. *Mucosal Immunol.* **7**, 818–828 (2014).
- Huch, M. & Koo, B.-K. Modeling mouse and human development using organoid cultures. *Development* **142**, 3113–3125 (2015).
- Zhao, Z. et al. Organoids. *Nat. Rev. Methods Prim.* **2**, 94 (2022).
- Kim, J., Koo, B. K. & Knoblich, J. A. Human organoids: model systems for human biology and medicine. *Nat. Rev. Mol. Cell Biol.* **21**, 571–584 (2020).
- Fowler, J. L., Ang, L. T. & Loh, K. M. A critical look: challenges in differentiating human pluripotent stem cells into desired cell types and organoids. *Wiley Interdiscip. Rev. Dev. Biol.* **9**, e368 (2020).
- van de Wetering, M. et al. Prospective derivation of a living organoid biobank of colorectal cancer patients. *Cell* **161**, 933–945 (2015).
- Sachs, N. et al. Long-term expanding human airway organoids for disease modeling. *EMBO J.* **38**, e100300 (2019).
- Huch, M. et al. Unlimited in vitro expansion of adult bi-potent pancreas progenitors through the Lgr5/R-spondin axis. *EMBO J.* **32**, 2708–2721 (2013).
- Jensen, K. B. & Little, M. H. Organoids are not organs: sources of variation and misinformation in organoid biology. *Stem Cell Rep.* **18**, 1255–1270 (2023).
- Rezania, A. et al. Reversal of diabetes with insulin-producing cells derived in vitro from human pluripotent stem cells. *Nat. Biotechnol.* **32**, 1121–1133 (2014).
- Veres, A. et al. Charting cellular identity during human in vitro β -cell differentiation. *Nature* **569**, 368–373 (2019).
- Beydag-Tasöz, B. S., Yennek, S. & Grapin-Botton, A. Towards a better understanding of diabetes mellitus using organoid models. *Nat. Rev. Endocrinol.* **19**, 232–248 (2023).
- Qian, X. et al. Brain-region-specific organoids using mini-bioreactors for modeling ZIKV exposure. *Cell* **165**, 1238–1254 (2016).
- Capowski, E. E. et al. Reproducibility and staging of 3D human retinal organoids across multiple pluripotent stem cell lines. *Development* **146**, dev171686 (2019).
- Gjorevski, N. et al. Tissue geometry drives deterministic organoid patterning. *Science* **375**, eaaw9021 (2022).
- Gjorevski et al. generate bioengineered organoids with an in vivo-like tissue architecture.**
- Liu, Q. et al. In situ vitrification of lung cancer organoids on a microwell array. *Micromachines* **12**, 624 (2021).
- Schuster, B. et al. Automated microfluidic platform for dynamic and combinatorial drug screening of tumor organoids. *Nat. Commun.* **11**, 5271 (2020).
- Dangles-Marie, V. et al. Establishment of human colon cancer cell lines from fresh tumors versus xenografts: comparison of success rate and cell line features. *Cancer Res.* **67**, 398–407 (2007).
- Fujii, M. et al. A colorectal tumor organoid library demonstrates progressive loss of niche factor requirements during tumorigenesis. *Cell Stem Cell* **18**, 827–838 (2016).
- Van Hemelryk, A. & van Weerden, W. M. Novel patient-derived 3D culture models to guide clinical decision-making in prostate cancer. *Curr. Opin. Endocr. Metab. Res.* **10**, 7–15 (2020).
- Kim, M. et al. Patient-derived lung cancer organoids as in vitro cancer models for therapeutic screening. *Nat. Commun.* **10**, 3991 (2019).
- Broutier, L. et al. Human primary liver cancer-derived organoid cultures for disease modeling and drug screening. *Nat. Med.* **23**, 1424–1435 (2017).
- Kleinman, H. K. & Martin, G. R. Matrigel: basement membrane matrix with biological activity. *Semin. Cancer Biol.* **15**, 378–386 (2005).
- Voog, J. & Jones, D. L. Stem cells and the niche: a dynamic duo. *Cell Stem Cell* **6**, 103–115 (2010).
- Clevers, H. The intestinal crypt, a prototype stem cell compartment. *Cell* **154**, 274–284 (2013).
- Velazquez, J. J. et al. Gene regulatory network analysis and engineering directs development and vascularization of multilineage human liver organoids. *Cell Syst.* **12**, 41–55.e11 (2021).
- Beumer, J. et al. High-resolution mRNA and secretome atlas of human enteroendocrine cells. *Cell* **181**, 1291–1306.e19 (2020).
- Azzarelli, R. et al. Neurogenin3 phosphorylation controls reprogramming efficiency of pancreatic ductal organoids into endocrine cells. *Sci. Rep.* **8**, 15374 (2018).
- Lee, J. et al. Expansion and conversion of human pancreatic ductal cells into insulin-secreting endocrine cells. *eLife* **2**, e00940 (2013).
- Beumer, J. & Clevers, H. Cell fate specification and differentiation in the adult mammalian intestine. *Nat. Rev. Mol. Cell Biol.* <https://doi.org/10.1038/s41580-020-0278-0> (2020).
- Cui, C., Wang, F., Zheng, Y., Wei, H. & Peng, J. From birth to death: the hardworking life of Paneth cell in the small intestine. *Front. Immunol.* **14**, 1122258 (2023).
- Gimeno, R. E., Briere, D. A. & Seeley, R. J. Leveraging the gut to treat metabolic disease. *Cell Metab.* **31**, 679–698 (2020).

67. Adolph, T. E. et al. Paneth cells as a site of origin for intestinal inflammation. *Nature* **503**, 272–276 (2013).
68. Stappenbeck, T. S. & McGovern, D. P. B. Paneth cell alterations in the development and phenotype of Crohn's disease. *Gastroenterology* **152**, 322–326 (2017).
69. Fujii, M. et al. Human intestinal organoids maintain self-renewal capacity and cellular diversity in niche-inspired culture condition. *Cell Stem Cell* **23**, 787–793.e6 (2018).
70. Mead, B. E. et al. Screening for modulators of the cellular composition of gut epithelia via organoid models of intestinal stem cell differentiation. *Nat. Biomed. Eng.* **6**, 476–494 (2022).
71. He, G.-W. et al. Optimized human intestinal organoid model reveals interleukin-22-dependency of Paneth cell formation. *Cell Stem Cell* **29**, 1333–1345.e6 (2022).
72. Eng, S. J. et al. FOXO inhibition rescues a-defensin expression in human intestinal organoids. *Proc. Natl Acad. Sci. USA* **120**, e2312453120 (2023).
73. Petersen, N. et al. Generation of L cells in mouse and human small intestine organoids. *Diabetes* **63**, 410–420 (2014).
74. Gehart, H. et al. Identification of enteroendocrine regulators by real-time single-cell differentiation mapping. *Cell* **176**, 1158–1173.e16 (2019).
75. Beumer, J. et al. Enteroendocrine cells switch hormone expression along the crypt-to-villus BMP signalling gradient. *Nat. Cell Biol.* **20**, 909–916 (2018).
76. Turco, M. Y. et al. Long-term, hormone-responsive organoid cultures of human endometrium in a chemically defined medium. *Nat. Cell Biol.* **19**, 568–577 (2017).
77. Boretto, M. et al. Development of organoids from mouse and human endometrium showing endometrial epithelium physiology and long-term expandability. *Development* **144**, 1775–1786 (2017).
78. Dong, Z. et al. Hypoxia promotes airway differentiation in the human lung epithelium. *Cell Stem Cell* **32**, 1705–1722.e9 (2025).
79. Burgess, C. L. et al. Generation of human alveolar epithelial type I cells from pluripotent stem cells. *Cell Stem Cell* **31**, 657–675.e8 (2024).
80. Lee, J. et al. Hair-bearing human skin generated entirely from pluripotent stem cells. *Nature* **582**, 399–404 (2020).
81. Hu, H. et al. Long-term expansion of functional mouse and human hepatocytes as 3D organoids. *Cell* **175**, 1591–1606.e19 (2018).
82. Andersson-Rolf, A. et al. Long-term in vitro expansion of a human fetal pancreas stem cell that generates all three pancreatic cell lineages. *Cell* **187**, 7394–7413.e22 (2024).
83. Birney, F. et al. Assembly of functionally integrated human forebrain spheroids. *Nature* **545**, 54–59 (2017).
84. Paşca, S. P. Assembling human brain organoids. *Science* **363**, 126–127 (2019).
85. Kim, J.-I. et al. Human assembloid model of the ascending neural sensory pathway. *Nature* **642**, 143–153 (2025).
86. Yuan, L. et al. Human assembloids recapitulate periportal liver tissue in vitro. *Nature* **650**, 438–449 (2026).
87. Koike, H. et al. Modelling human hepato-biliary-pancreatic organogenesis from the foregut-midgut boundary. *Nature* **574**, 112–116 (2019).
88. Co, J. Y., Margalef-Català, M., Monack, D. M. & Amieva, M. R. Controlling the polarity of human gastrointestinal organoids to investigate epithelial biology and infectious diseases. *Nat. Protoc.* **16**, 5171–5192 (2021).
89. Stroulios, G. et al. Apical-out airway organoids as a platform for studying viral infections and screening for antiviral drugs. *Sci. Rep.* **12**, 7673 (2022).
90. Kakni, P., López-Iglesias, C., Truckenmüller, R., Habibović, P. & Giselbrecht, S. PSC-derived intestinal organoids with apical-out orientation as a tool to study nutrient uptake, drug absorption and metabolism. *Front. Mol. Biosci.* **10**, 1102209 (2023).
91. Co, J. Y. et al. Controlling epithelial polarity: a human enteroid model for host-pathogen interactions. *Cell Rep.* **26**, 2509–2520.e4 (2019).
92. Chiu, M. C. et al. Apical-out human airway organoids modeling SARS-CoV-2 infection. *Viruses* **15**, 1166 (2023).
93. Dahmann, C., Oates, A. C. & Brand, M. Boundary formation and maintenance in tissue development. *Nat. Rev. Genet.* **12**, 43–55 (2011).
94. Wang, Y. et al. A microengineered collagen scaffold for generating a polarized crypt-villus architecture of human small intestinal epithelium. *Biomaterials* **128**, 44–55 (2017).
95. Kasendra, M. et al. Development of a primary human Small Intestine-on-a-Chip using biopsy-derived organoids. *Sci. Rep.* **8**, 2871 (2018).
96. Nikolaev, M. et al. Homeostatic mini-intestines through scaffold-guided organoid morphogenesis. *Nature* **585**, 574–578 (2020).
97. Gjorevski, N. et al. Designer matrices for intestinal stem cell and organoid culture. *Nature* **539**, 560–564 (2016).
98. Murphy, S. V. & Atala, A. 3D bioprinting of tissues and organs. *Nat. Biotechnol.* **32**, 773–785 (2014).
99. Brassard, J. A., Nikolaev, M., Hübscher, T., Hofer, M. & Lutolf, M. P. Recapitulating macro-scale tissue self-organization through organoid bioprinting. *Nat. Mater.* **20**, 22–29 (2021).
100. Roth, J. G. et al. Spatially controlled construction of assembloids using bioprinting. *Nat. Commun.* **14**, 4346 (2023).
101. Kim, E. et al. Creation of bladder assembloids mimicking tissue regeneration and cancer. *Nature* **588**, 664–669 (2020).
102. Lawlor, K. T. et al. Cellular extrusion bioprinting improves kidney organoid reproducibility and conformation. *Nat. Mater.* **20**, 260–271 (2021).
103. Zhang, Y. et al. Biomimetic niches reveal the minimal cues to trigger apical lumen formation in single hepatocytes. *Nat. Mater.* **19**, 1026–1035 (2020).
104. Wang, Y. et al. Nature-inspired micropatterns. *Nat. Rev. Methods Prim.* **3**, 1–22 (2023).
105. Hu, X. & Bao, M. Advances in micropatterning technology for mechanotransduction research. *Mechanobiol. Med.* **2**, 100066 (2024).
106. Karzbrun, E. et al. Human neural tube morphogenesis in vitro by geometric constraints. *Nature* **599**, 268–272 (2021).
107. Vining, K. H. & Mooney, D. J. Mechanical forces direct stem cell behaviour in development and regeneration. *Nat. Rev. Mol. Cell Biol.* **18**, 728–742 (2017).
108. Xi, W., Saw, T. B., Delacour, D., Lim, C. T. & Ladoux, B. Material approaches to active tissue mechanics. *Nat. Rev. Mater.* **4**, 23–44 (2018).
109. Sorrentino, G. et al. Mechano-modulatory synthetic niches for liver organoid derivation. *Nat. Commun.* **11**, 3416 (2020).
110. Voges, H. K. et al. Development of a human cardiac organoid injury model reveals innate regenerative potential. *Development* **144**, 1118–1127 (2017).
111. Yang, X., Pabon, L. & Murry, C. E. Engineering adolescence: maturation of human pluripotent stem cell-derived cardiomyocytes. *Circ. Res.* **114**, 511–523 (2014).
112. Abdel Fattah, A. R. et al. Actuation enhances patterning in human neural tube organoids. *Nat. Commun.* **12**, 3192 (2021).
113. Yang, Q. et al. Cell fate coordinates mechano-osmotic forces in intestinal crypt formation. *Nat. Cell Biol.* **23**, 733–744 (2021).
114. Huang, D. et al. Reversed-engineered human alveolar lung-on-a-chip model. *Proc. Natl Acad. Sci. USA* **118**, e2016146118 (2021).
115. Zamprogno, P. et al. Second-generation lung-on-a-chip with an array of stretchable alveoli made with a biological membrane. *Commun. Biol.* **4**, 168 (2021).
116. Huh, D. et al. Reconstituting organ-level lung functions on a chip. *Science* **328**, 1662–1668 (2010).
117. Leiby, K. L. et al. Rational engineering of lung alveolar epithelium. *npj Regen. Med.* **8**, 22 (2023).
118. Sharma, K. et al. Dynamic persistence of UPEC intracellular bacterial communities in a human bladder-chip model of urinary tract infection. *eLife* **10**, e66481 (2021).
119. Chiu, M. C. et al. A bipotential organoid model of respiratory epithelium recapitulates high infectivity of SARS-CoV-2 Omicron variant. *Cell Discov.* **8**, 57 (2022).
120. Abo, K. M. et al. Air-liquid interface culture promotes maturation and allows environmental exposure of pluripotent stem cell-derived alveolar epithelium. *JCI Insight* **7**, e155589 (2022).
121. Kloxin, A. M., Kasko, A. M., Salinas, C. N. & Anseth, K. S. Photodegradable hydrogels for dynamic tuning of physical and chemical properties. *Science* **324**, 59–63 (2009).
122. Ruskowitz, E. R. & DeForest, C. A. Photoresponsive biomaterials for targeted drug delivery and 4D cell culture. *Nat. Rev. Mater.* **3**, 1–17 (2018).
123. Lee, K. K. et al. Human stomach-on-a-chip with luminal flow and peristaltic-like motility. *Lab Chip* **18**, 3079–3085 (2018).
124. Tao, T. et al. Engineering human islet organoids from iPSCs using an organ-on-chip platform. *Lab Chip* **19**, 948–958 (2019).
125. Zheng, Y. et al. Controlled modelling of human epiblast and amnion development using stem cells. *Nature* **573**, 421–425 (2019).
126. Ingber, D. E. Human organs-on-chips for disease modelling, drug development and personalized medicine. *Nat. Rev. Genet.* **23**, 467–491 (2022).
127. Cakir, B. et al. Engineering of human brain organoids with a functional vascular-like system. *Nat. Methods* **16**, 1169–1175 (2019).
128. Takebe, T. et al. Vascularized and complex organ buds from diverse tissues via mesenchymal cell-driven condensation. *Cell Stem Cell* **16**, 556–565 (2015).
129. Homan, K. A. et al. Flow-enhanced vascularization and maturation of kidney organoids in vitro. *Nat. Methods* **16**, 255–262 (2019).
130. Koledova, Z. & Lu, P. A 3D fibroblast-epithelium co-culture model for understanding microenvironmental role in branching morphogenesis of the mammary gland. *Methods Mol. Biol.* **1501**, 217–231 (2017).
131. Jowett, G. M. et al. Organoids capture tissue-specific innate lymphoid cell development in mice and humans. *Cell Rep.* **40**, 111281 (2022).
132. Noel, G. et al. A primary human macrophage-enteroid co-culture model to investigate mucosal gut physiology and host-pathogen interactions. *Sci. Rep.* **7**, 45270 (2017).
133. Tominaga, K. et al. Generation of human intestinal organoids containing tissue-resident immune cells. *Gastroenterology* **162**, S57 (2022).
134. Bouffi, C. et al. In vivo development of immune tissue in human intestinal organoids transplanted into humanized mice. *Nat. Biotechnol.* **41**, 824–831 (2023).
135. Lee, J.-H. et al. Lung stem cell differentiation in mice directed by endothelial cells via a BMP4-NFATc1-thrombospondin-1 axis. *Cell* **156**, 440–455 (2014).
136. Richards, Z. et al. Prostate stroma increases the viability and maintains the branching phenotype of human prostate organoids. *iScience* **12**, 304–317 (2019).
137. Schreurs, R. R. C. E. et al. Human fetal TNF- α -cytokine-producing CD4⁺ effector memory T cells promote intestinal development and mediate inflammation early in life. *Immunity* **50**, 462–476.e8 (2019).
138. Recaladin, T. et al. Human organoids with an autologous tissue-resident immune compartment. *Nature* **633**, 165–173 (2024).
139. Pacesa, M., Pelea, O. & Jinek, M. Past, present, and future of CRISPR genome editing technologies. *Cell* **187**, 1076–1100 (2024).
140. Geurts, M. H. & Clevers, H. CRISPR engineering in organoids for gene repair and disease modelling. *Nat. Rev. Bioeng.* **1**, 32–45 (2023).
141. Teriyapirom, I., Batista-Rocha, A. S. & Koo, B.-K. Genetic engineering in organoids. *J. Mol. Med.* **99**, 555–568 (2021).

142. van Rijn, J. M. et al. Intestinal failure and aberrant lipid metabolism in patients with DGAT1 deficiency. *Gastroenterology* **155**, 130–143.e15 (2018).
143. Hendriks, D. et al. Engineered human hepatocyte organoids enable CRISPR-based target discovery and drug screening for steatosis. *Nat. Biotechnol.* <https://doi.org/10.1038/s41587-023-01680-4> (2023).
144. Schwank, G. et al. Functional repair of CFTR by CRISPR/Cas9 in intestinal stem cell organoids of cystic fibrosis patients. *Cell Stem Cell* **13**, 653–658 (2013).
This paper presents gene correction in TSC-derived organoids.
145. Geurts, M. H. et al. CRISPR-based adenine editors correct nonsense mutations in a cystic fibrosis organoid biobank. *Cell Stem Cell* **26**, 503–510.e7 (2020).
146. Ruan, J. et al. Efficient gene editing at major CFTR mutation loci. *Mol. Ther. Nucleic Acids* **16**, 73–81 (2019).
147. Vishy, C. E. et al. Genetics of cystogenesis in base-edited human organoids reveal therapeutic strategies for polycystic kidney disease. *Cell Stem Cell* **31**, 537–553.e5 (2024).
148. Wu, Y. et al. Robust and inducible genome editing via an all-in-one prime editor in human pluripotent stem cells. *Nat. Commun.* **15**, 10824 (2024).
149. Xu, Y. et al. Adult human kidney organoids originate from CD24⁺ cells and represent an advanced model for adult polycystic kidney disease. *Nat. Genet.* **54**, 1690–1701 (2022).
150. Drost, J. et al. Sequential cancer mutations in cultured human intestinal stem cells. *Nature* **521**, 43–47 (2015).
151. Matano, M. et al. Modeling colorectal cancer using CRISPR-Cas9-mediated engineering of human intestinal organoids. *Nat. Med.* **21**, 256–262 (2015).
Together with Drost et al. (2015), this paper provides examples of using CRISPR to model tumorigenesis.
152. Drost, J. et al. Use of CRISPR-modified human stem cell organoids to study the origin of mutational signatures in cancer. *Science* **358**, 234–238 (2017).
153. Koo, B.-K. et al. Tumour suppressor RNF43 is a stem-cell E3 ligase that induces endocytosis of Wnt receptors. *Nature* **488**, 665–669 (2012).
154. Yan, H. H. N. et al. Organoid cultures of early-onset colorectal cancers reveal distinct and rare genetic profiles. *Gut* **69**, 2165–2179 (2020).
155. Jager, M. et al. Deficiency of nucleotide excision repair is associated with mutational signature observed in cancer. *Genome Res.* **29**, 1067–1077 (2019).
156. Kawasaki, K. et al. Chromosome engineering of human colon-derived organoids to develop a model of traditional serrated adenoma. *Gastroenterology* **158**, 638–651.e8 (2020).
157. Kawasaki, K. et al. An organoid biobank of neuroendocrine neoplasms enables genotype-phenotype mapping. *Cell* **183**, 1420–1435.e21 (2020).
158. Geurts, M. H. et al. One-step generation of tumor models by base editor multiplexing in adult stem cell-derived organoids. *Nat. Commun.* **14**, 4998 (2023).
159. Stolze, B., Reinhardt, S., Bullinger, L., Fröhling, S. & Scholl, C. Comparative analysis of KRAS codon 12, 13, 18, 61, and 117 mutations using human MCF10A isogenic cell lines. *Sci. Rep.* **5**, 8535 (2015).
160. Fumagalli, A. et al. Genetic dissection of colorectal cancer progression by orthotopic transplantation of engineered cancer organoids. *Proc. Natl Acad. Sci. USA* **114**, E2357–E2364 (2017).
161. Artegiani, B. et al. Probing the tumor suppressor function of BAP1 in CRISPR-engineered human liver organoids. *Cell Stem Cell* **24**, 927–943.e6 (2019).
162. Seino, T. et al. Human pancreatic tumor organoids reveal loss of stem cell niche factor dependence during disease progression. *Cell Stem Cell* **22**, 454–467.e6 (2018).
163. Michels, B. E. et al. Pooled in vitro and in vivo CRISPR-Cas9 screening identifies tumor suppressors in human colon organoids. *Cell Stem Cell* **26**, 782–792.e7 (2020).
164. Ringel, T. et al. Genome-scale CRISPR screening in human intestinal organoids identifies drivers of TGF- β resistance. *Cell Stem Cell* **26**, 431–440.e8 (2020).
165. Lin, L. et al. Unbiased transcription factor CRISPR screen identifies ZNF800 as master repressor of enteroendocrine differentiation. *Science* **382**, 451–458 (2023).
166. Hansen, S. L. et al. An organoid-based CRISPR-Cas9 screen for regulators of intestinal epithelial maturation and cell fate. *Sci. Adv.* **9**, eadg4055 (2023).
167. Sun, D. et al. SOX9 maintains human foetal lung tip progenitor state by enhancing WNT and RTK signalling. *EMBO J.* **41**, e111338 (2022).
168. Papp, D., Korcsmáros, T. & Hautefort, I. Revolutionizing immune research with organoid-based co-culture and chip systems. *Clin. Exp. Immunol.* <https://doi.org/10.1093/cei/uxae004> (2024).
169. Subtil, B. et al. Dendritic cell phenotype and function in a 3D co-culture model of patient-derived metastatic colorectal cancer organoids. *Front. Immunol.* **14**, 1105244 (2023).
170. Fang, H. et al. SIRT1 induces the accumulation of TAMs at colorectal cancer tumor sites via the CXCR4/CXCL12 axis. *Cell. Immunol.* **371**, 104458 (2022).
171. Marcon, F. et al. NK cells in pancreatic cancer demonstrate impaired cytotoxicity and a regulatory IL-10 phenotype. *Oncoimmunology* **9**, 1845424 (2020).
172. Koh, V. et al. Hedgehog transcriptional effector GLI mediates mTOR-induced PD-L1 expression in gastric cancer organoids. *Cancer Lett.* **518**, 59–71 (2021).
173. Holokai, L. et al. Murine- and human-derived autologous organoid/immune cell co-cultures as pre-clinical models of pancreatic ductal adenocarcinoma. *Cancers* **12**, 3816 (2020).
174. Eicher, A. K. et al. Functional human gastrointestinal organoids can be engineered from three primary germ layers derived separately from pluripotent stem cells. *Cell Stem Cell* **29**, 36–51.e6 (2022).
175. Naruse, M. et al. Re-expression of REG family and DUOXs genes in CRC organoids by co-culturing with CAFs. *Sci. Rep.* **11**, 2077 (2021).
176. Schuth, S. et al. Patient-specific modeling of stroma-mediated chemoresistance of pancreatic cancer using a three-dimensional organoid-fibroblast co-culture system. *J. Exp. Clin. Cancer Res.* **41**, 312 (2022).
177. Chang, J. T. Pathophysiology of inflammatory bowel diseases. *N. Engl. J. Med.* **383**, 2652–2664 (2020).
178. Baumgart, D. C. & Carding, S. R. Inflammatory bowel disease: cause and immunobiology. *Lancet* **369**, 1627–1640 (2007).
179. Pavlidis, P. et al. Cytokine responsive networks in human colonic epithelial organoids unveil a molecular classification of inflammatory bowel disease. *Cell Rep.* **40**, 111439 (2022).
180. Hibiya, S. et al. Long-term inflammation transforms intestinal epithelial cells of colonic organoids. *J. Crohns Colitis* **11**, 621–630 (2017).
181. Takashima, S. et al. T cell-derived interferon- γ programs stem cell death in immune-mediated intestinal damage. *Sci. Immunol.* **4**, eaay8556 (2019).
182. Hammoudi, N. et al. Autologous organoid co-culture model reveals T cell-driven epithelial cell death in Crohn's disease. *Front. Immunol.* **13**, 1008456 (2022).
183. Beauvillage, C. et al. Development of a human primary gut-on-a-chip to model inflammatory processes. *Sci. Rep.* **10**, 1–16 (2020).
184. Roh, T. T., Chen, Y., Paul, H. T., Guo, C. & Kaplan, D. L. 3D bioengineered tissue model of the large intestine to study inflammatory bowel disease. *Biomaterials* **225**, 119517 (2019).
185. Engevik, M. A. et al. Human *Clostridium difficile* infection: inhibition of NHE3 and microbiota profile. *Am. J. Physiol. Gastrointest. Liver Physiol.* **308**, G497–G509 (2015).
186. Hou, Q. et al. *Lactobacillus* accelerates ISCs regeneration to protect the integrity of intestinal mucosa through activation of STAT3 signaling pathway induced by LPLs secretion of IL-22. *Cell Death Differ.* **25**, 1657–1670 (2018).
187. Jowett, G. M., Coales, I. & Neves, J. F. Organoids as a tool for understanding immune-mediated intestinal regeneration and development. *Development* **149**, dev199904 (2022).
188. Brice, D. P. et al. Interleukin-27 regulates the function of the gastrointestinal epithelial barrier in a human tissue-derived organoid model. *Biology* **11**, 427 (2022).
189. Lindemans, C. A. et al. Interleukin-22 promotes intestinal-stem-cell-mediated epithelial regeneration. *Nature* **528**, 560–564 (2015).
190. Keir, M., Yi, T., Lu, T. & Ghilardi, N. The role of IL-22 in intestinal health and disease. *J. Exp. Med.* **217**, e20192195 (2020).
191. Rothenberg, M. E. et al. Randomized phase I healthy volunteer study of UTTR1147A (IL-22Fc): a potential therapy for epithelial injury. *Clin. Pharmacol. Ther.* **105**, 177–189 (2019).
192. Yui, S. et al. YAP/TAZ-dependent reprogramming of colonic epithelium links ECM remodeling to tissue regeneration. *Cell Stem Cell* **22**, 35–49.e7 (2018).
193. Aloia, L. et al. Epigenetic remodelling licenses adult cholangiocytes for organoid formation and liver regeneration. *Nat. Cell Biol.* **21**, 1321–1333 (2019).
194. Choi, J. et al. Release of Notch activity coordinated by IL-1 β signalling confers differentiation plasticity of airway progenitors via Fosl2 during alveolar regeneration. *Nat. Cell Biol.* **23**, 953–966 (2021).
195. Sugimoto, S. et al. Reconstruction of the human colon epithelium in vivo. *Cell Stem Cell* **22**, 171–176.e5 (2018).
196. Bannier-Hélaouët, M. et al. Human conjunctiva organoids to study ocular surface homeostasis and disease. *Cell Stem Cell* **31**, 227–243.e12 (2024).
197. Soto-Gomez, A., Gunawan, J. P., Barazzuol, L., Pringle, S. & Coppes, R. P. Organoid-based personalized medicine: from tumor outcome prediction to autologous transplantation. *Stem Cells* <https://doi.org/10.1093/stemcells/sxae023> (2024).
198. Loomans, C. J. M. et al. Expansion of adult human pancreatic tissue yields organoids harboring progenitor cells with endocrine differentiation potential. *Stem Cell Rep.* **10**, 712–724 (2018).
199. Lee, H. J. et al. Significance of CD133 as a cancer stem cell markers focusing on the tumorigenicity of pancreatic cancer cell lines. *J. Korean Surg. Soc.* **81**, 263–270 (2011).
200. Clevers, H. et al. Tissue-engineering the intestine: the trials before the trials. *Cell Stem Cell* **24**, 855–859 (2019).
201. Sugimoto, S. et al. An organoid-based organ-repurposing approach to treat short bowel syndrome. *Nature* **592**, 99–104 (2021).
202. van Zanten, J. et al. Optimization of the production process of clinical-grade human salivary gland organoid-derived cell therapy for the treatment of radiation-induced xerostomia in head and neck cancer. *Pharmaceutics* **16**, 435 (2024).
203. de Klerk, E. & Hebrok, M. Stem cell-based clinical trials for diabetes mellitus. *Front. Endocrinol.* **12**, 631463 (2021).
204. US National Library of Medicine. *ClinicalTrials.gov* <https://clinicaltrials.gov/study/NCT04786262> (2026).
205. Algemeen Nederlands Persbureau. First treatment in the Netherlands: cancer patient treated with stem cells from own salivary gland. *NL Times* <https://nltimes.nl/2022/12/18/first-treatment-netherlands-cancer-patient-treated-stem-cells-salivary-gland/> (18 December 2022).
206. Fogel, D. B. Factors associated with clinical trials that fail and opportunities for improving the likelihood of success: a review. *Contemp. Clin. Trials Commun.* **11**, 156–164 (2018).
207. Menche, C. & Farin, H. F. Strategies for genetic manipulation of adult stem cell-derived organoids. *Exp. Mol. Med.* **53**, 1483–1494 (2021).
208. Sachs, N. et al. A living biobank of breast cancer organoids captures disease heterogeneity. *Cell* **172**, 373–386.e10 (2018).
209. Tebon, P. J. et al. Drug screening at single-organoid resolution via bioprinting and interferometry. *Nat. Commun.* **14**, 3168 (2023).

210. Maloney, E. et al. Immersion bioprinting of tumor organoids in multi-well plates for increasing chemotherapy screening throughput. *Micromachines* **11**, 208 (2020).
211. Calandrin, C. et al. Organoid-based drug screening reveals neddylation as therapeutic target for malignant rhabdoid tumors. *Cell Rep.* **36**, 109568 (2021).
212. Driehuis, E. et al. Pancreatic cancer organoids recapitulate disease and allow personalized drug screening. *Proc. Natl Acad. Sci. USA* **116**, 26580–26590 (2019).
213. Verissimo, C. S. et al. Targeting mutant RAS in patient-derived colorectal cancer organoids by combinatorial drug screening. *eLife* **5**, e18489 (2016).
214. Dayton, T. L. et al. Druggable growth dependencies and tumor evolution analysis in patient-derived organoids of neuroendocrine neoplasms from multiple body sites. *Cancer Cell* **41**, 2083–2099.e9 (2023).
215. Al Shihabi, A. et al. Personalized chordoma organoids for drug discovery studies. *Sci. Adv.* **8**, eabl3674 (2022).
216. Hirt, C. K. et al. Drug screening and genome editing in human pancreatic cancer organoids identifies drug–gene interactions and candidates for off-label treatment. *Cell Genom.* **2**, 100095 (2022).
217. Mertens, S. et al. Drug-repurposing screen on patient-derived organoids identifies therapy-induced vulnerability in KRAS-mutant colon cancer. *Cell Rep.* **42**, 112324 (2023).
218. Smabers, L. P. et al. Organoids as a biomarker for personalized treatment in metastatic colorectal cancer: drug screen optimization and correlation with patient response. *J. Exp. Clin. Cancer Res.* **43**, 61 (2024).
219. de Poel, E. et al. FDA-approved drug screening in patient-derived organoids demonstrates potential of drug repurposing for rare cystic fibrosis genotypes. *J. Cyst. Fibros.* **22**, 548–559 (2023).
220. Tindle, C. et al. A living organoid biobank of patients with Crohn's disease reveals molecular subtypes for personalized therapeutics. *Cell Rep. Med.* **5**, 101748 (2024).
221. Sarvestani, S. K. et al. Induced organoids derived from patients with ulcerative colitis recapitulate colonic reactivity. *Nat. Commun.* **12**, 262 (2021).
222. Niklinska-Schirtz, B. J. et al. Ileal derived organoids from Crohn's disease patients show unique transcriptomic and secretomic signatures. *Cell Mol. Gastroenterol. Hepatol.* **12**, 1267–1280 (2021).
223. Yan, H. H. N. et al. A comprehensive human gastric cancer organoid biobank captures tumor subtype heterogeneity and enables therapeutic screening. *Cell Stem Cell* **23**, 882–897.e11 (2018).
224. Millen, R. et al. Patient-derived head and neck cancer organoids allow treatment stratification and serve as a tool for biomarker validation and identification. *Med* **4**, 290–310.e12 (2023).
225. Chalabi, M. et al. Neoadjuvant immunotherapy leads to pathological responses in MMR-proficient and MMR-deficient early-stage colon cancers. *Nat. Med.* **26**, 566–576 (2020).
226. Yao, Y. et al. Patient-derived organoids predict chemoradiation responses of locally advanced rectal cancer. *Cell Stem Cell* **26**, 17–26.e6 (2020).
227. Vlachogiannis, G. et al. Patient-derived organoids model treatment response of metastatic gastrointestinal cancers. *Science* **359**, 920–926 (2018).
228. Ganesh, K. et al. A rectal cancer organoid platform to study individual responses to chemoradiation. *Nat. Med.* **25**, 1607–1614 (2019).
229. Wensink, G. E. et al. Patient-derived organoids as a predictive biomarker for treatment response in cancer patients. *npj Precis. Oncol.* **5**, 30 (2021).
230. de Witte, C. J. et al. Patient-derived ovarian cancer organoids mimic clinical response and exhibit heterogeneous inter- and inpatient drug responses. *Cell Rep.* **31**, 107762 (2020).
231. Mazzocchi, A. R., Rajan, S. A. P., Votanopoulos, K. I., Hall, A. R. & Skardal, A. In vitro patient-derived 3D mesothelioma tumor organoids facilitate patient-centric therapeutic screening. *Sci. Rep.* **8**, 2886 (2018).
232. Mo, S. et al. Patient-derived organoids from colorectal cancer with paired liver metastasis reveal tumor heterogeneity and predict response to chemotherapy. *Adv. Sci.* **9**, e2204097 (2022).
233. Ooft, S. N. et al. Patient-derived organoids can predict response to chemotherapy in metastatic colorectal cancer patients. *Sci. Transl. Med.* **11**, eaay2574 (2019).
234. Liu, J. et al. Cancer-associated fibroblasts provide a stromal niche for liver cancer organoids that confers trophic effects and therapy resistance. *Cell Mol. Gastroenterol. Hepatol.* **11**, 407–431 (2021).
235. Farin, H. F. et al. Colorectal cancer organoid-stroma biobank allows subtype-specific assessment of individualized therapy responses. *Cancer Discov.* **13**, 2192–2211 (2023).
236. Kokkios, J. et al. Ex vivo culture of intact human patient derived pancreatic tumour tissue. *Sci. Rep.* **11**, 1944 (2021).
237. Voabil, P. et al. An ex vivo tumor fragment platform to dissect response to PD-1 blockade in cancer. *Nat. Med.* **27**, 1250–1261 (2021).
238. Misra, S. et al. Ex vivo organotypic culture system of precision-cut slices of human pancreatic ductal adenocarcinoma. *Sci. Rep.* **9**, 2133 (2019).
239. Ding, S. et al. Patient-derived micro-organospheres enable clinical precision oncology. *Cell Stem Cell* **29**, 905–917.e6 (2022).
240. Yang, H. et al. Pharmacogenomic profiling of intra-tumor heterogeneity using a large organoid biobank of liver cancer. *Cancer Cell* **42**, 535–551.e8 (2024).
241. Zhang, J. et al. Immunotherapy discovery on tumor organoid-on-a-chip platforms that recapitulate the tumor microenvironment. *Adv. Drug Deliv. Rev.* **187**, 114365 (2022).
242. Magré, L. et al. Emerging organoid-immune co-culture models for cancer research: from oncoimmunology to personalized immunotherapies. *J. Immunother. Cancer* **11**, e006290 (2023).
243. Justiz Vaillant, A. A., Nessel, T. A. & Zito, P. M. *Immunotherapy* (StatPearls, 2023).
244. Dao, V., Yuki, K., Lo, Y.-H., Nakano, M. & Kuo, C. J. Immune organoids: from tumor modeling to precision oncology. *Trends Cancer Res.* **8**, 870–880 (2022).
245. Knoblauch, M. et al. In-vitro model to mimic T cell subset change in human PDAC organoid co-culture. *J. Cancer Res. Clin. Oncol.* <https://doi.org/10.1007/s00432-023-05100-7> (2023).
246. Neal, J. T. et al. Organoid modeling of the tumor immune microenvironment. *Cell* **175**, 1972–1988.e16 (2018).
247. Yu, L. et al. Patient-derived organoids of bladder cancer recapitulate antigen expression profiles and serve as a personal evaluation model for CAR-T cells in vitro. *Clin. Transl. Immunol.* **10**, e1248 (2021).
248. Zhou, G. et al. Modelling immune cytotoxicity for cholangiocarcinoma with tumour-derived organoids and effector T cells. *Br. J. Cancer* **127**, 649–660 (2022).
249. Schnalzger, T. E. et al. 3D model for CAR-mediated cytotoxicity using patient-derived colorectal cancer organoids. *EMBO J.* **38**, e100928 (2019).
250. Harter, M. F. et al. Analysis of off-tumour toxicities of T-cell-engaging bispecific antibodies via donor-matched intestinal organoids and tumouroids. *Nat. Biomed. Eng.* <https://doi.org/10.1038/s41551-023-01156-5> (2023).
251. Sun, D., Gao, W., Hu, H. & Zhou, S. Why 90% of clinical drug development fails and how to improve it? *Acta Pharm. Sin.* **43**, 3049–3062 (2022).
252. Martignoni, M., Groothuis, G. M. M. & de Kanter, R. Species differences between mouse, rat, dog, monkey and human CYP-mediated drug metabolism, inhibition and induction. *Expert Opin. Drug Metab. Toxicol.* **2**, 875–894 (2006).
253. Ha, C. E. & Bhagavan, N. V. in *Essentials of Medical Biochemistry* 3rd edn (eds Ha, C. E. & Bhagavan, N. V.) 795–805 (Academic, 2023).
254. Zietek, T., Rath, E., Haller, D. & Daniel, H. Intestinal organoids for assessing nutrient transport, sensing and incretin secretion. *Sci. Rep.* **5**, 1–10 (2015).
255. Zietek, T. et al. Organoids to study intestinal nutrient transport, drug uptake and metabolism — update to the human model and expansion of applications. *Front. Bioeng. Biotechnol.* **8**, 577656 (2020).
256. Kasendra, M. et al. Duodenum intestine-chip for preclinical drug assessment in a human relevant model. *eLife* **9**, e50135 (2020).
257. Park, E. et al. Development of organoid-based drug metabolism model. *Toxicol. Appl. Pharmacol.* **385**, 114790 (2019).
258. Inui, T. et al. Functional intestinal monolayers from organoids derived from human iPSC cells for drug discovery research. *Stem Cell Res. Ther.* **15**, 57 (2024).
259. Zhang, C. J. et al. A human liver organoid screening platform for DILI risk prediction. *J. Hepatol.* **78**, 998–1006 (2023).
260. Bouwmeester, M. C. et al. Drug metabolism of hepatocyte-like organoids and their applicability in in vitro toxicity testing. *Molecules* **28**, 621 (2023).
261. Aceves, J. O. et al. 3D proximal tubule-on-chip model derived from kidney organoids with improved drug uptake. *Sci. Rep.* **12**, 14997 (2022).
262. Susa, K. et al. ATP/ADP biosensor organoids for drug nephrotoxicity assessment. *Front. Cell Dev. Biol.* **11**, 1138504 (2023).
263. Rizki-Safitri, A. et al. Live functional assays reveal longitudinal maturation of transepithelial transport in kidney organoids. *Front. Cell Dev. Biol.* **10**, 978888 (2022).
264. Kim, H. et al. Diverse bat organoids provide pathophysiological models for zoonotic viruses. *Science* **388**, 756–762 (2025).
265. Takebe, T. et al. Vascularized and functional human liver from an iPSC-derived organ bud transplant. *Nature* **499**, 481–484 (2013).
- Takebe et al. describe one of the first examples of the assembly approach to generate liver bud organoids.**
266. Dowbaj, A. M. et al. Mouse liver assembloids model periportal architecture and biliary fibrosis. *Nature* **644**, 473–482 (2025).
- This research starts with hTSC hepatocyte organoids to form assembloids that recapitulate the architecture of the liver periportal region.**
267. Yang, L. et al. Innate immune signaling in trophoblast and decidua organoids defines differential antiviral defenses at the maternal–fetal interface. *eLife* **11**, 1e79794 (2022).
268. Lorenzo-Martin, L. F. et al. Spatiotemporally resolved colorectal oncogenesis in mini-colons ex vivo. *Nature* <https://doi.org/10.1038/s41586-024-07330-2> (2024).
269. Abramson, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493–500 (2024).
270. Motmaen, A. et al. Peptide-binding specificity prediction using fine-tuned protein structure prediction networks. *Proc. Natl Acad. Sci. USA* **120**, e2216697120 (2023).
271. Yu, L. et al. Blastocyst-like structures generated from human pluripotent stem cells. *Nature* **591**, 620–626 (2021).
272. Kagawa, H. et al. Human blastoids model blastocyst development and implantation. *Nature* **601**, 600–605 (2022).
273. Yanagida, A. et al. Naive stem cell blastocyst model captures human embryo lineage segregation. *Cell Stem Cell* **28**, 1016–1022.e4 (2021).
274. Moris, N. et al. An in vitro model of early anteroposterior organization during human development. *Nature* **582**, 410–415 (2020).
275. Hamazaki, N. et al. Retinoic acid induces human gastruloids with posterior embryo-like structures. *Nat. Cell Biol.* **26**, 1790–1803 (2024).
276. Miao, Y. et al. Reconstruction and deconstruction of human somitogenesis in vitro. *Nature* **614**, 500–508 (2023).
277. Yamanaka, Y. et al. Reconstituting human somitogenesis in vitro. *Nature* **614**, 509–520 (2023).

278. Cugola, F. R. et al. The Brazilian Zika virus strain causes birth defects in experimental models. *Nature* **534**, 267–271 (2016).
279. Xu, M. et al. Identification of small-molecule inhibitors of Zika virus infection and induced neural cell death via a drug repurposing screen. *Nat. Med.* **22**, 1101–1107 (2016).
280. Yoon, K.-J. et al. Zika-virus-encoded NS2A disrupts mammalian cortical neurogenesis by degrading adherens junction proteins. *Cell Stem Cell* **21**, 349–358.e6 (2017).
281. Schmidt, C. et al. Multi-chamber cardioids unravel human heart development and cardiac defects. *Cell* **186**, 5587–5605.e27 (2023).
282. Casamitjana, J., Espinet, E. & Rovira, M. Pancreatic organoids for regenerative medicine and cancer research. *Front. Cell Dev. Biol.* **10**, 886153 (2022).
283. Gerli, M. F. M. et al. Single-cell guided prenatal derivation of primary fetal epithelial organoids from human amniotic and tracheal fluids. *Nat. Med.* **30**, 875–887 (2024).
284. He, P. et al. A human fetal lung cell atlas uncovers proximal-distal gradients of differentiation and key regulators of epithelial fates. *Cell* **185**, 4841–4860.e25 (2022).
285. Guiu, J. et al. Tracing the origin of adult intestinal stem cells. *Nature* **570**, 107–111 (2019).

Acknowledgements

The authors thank T. Grenier for helpful comments and discussion and are very grateful to the Mendjan, Huch and Gjorevski laboratory for providing images. The authors also thank the reviewers for the constructive comments and apologize for not being able to cite all relevant papers.

Author contributions

A.A.-R. wrote the first draft of the manuscript. A.A.-R and H.C. were involved in the discussion of content and at all stages of revision.

Competing interests

H.C. is a member of the scientific advisory board of Cell. H.C. is an inventor on multiple patents related to organoid technology. His full disclosure is available given at <http://www.uu.nl/staff/JCClevers/Additionalfunctions>. H.C. currently holds no advisorships to commercial entities but holds stock in Roche, in D1Med (Shanghai), in Xilis and in Surrozen. A.A.-R declares no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41580-026-00974-0>.

Peer review information *Nature Reviews Molecular Cell Biology* thanks Gordana Vunjak-Novakovic and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© Springer Nature Limited 2026